

RTL Design Sherpa

amber — Blocking MESI Snoopy L1 Cache Micro-Architecture Specification 0.5

October 6, 2026

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List of Waveforms

No waveforms in this document.

1 Amber Mas Index

Generated: 2026-10-15

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2 amber — Blocking MESI Snoopy L1 Cache — Hardware Architecture Specification

Version: 0.9 **Date:** 2026-10-11 **Status:** full HAS ratifies the Pre-HAS sketch; book-informed pass folded Chinnakonda multiprocessor-cache material into coherence FSM, snoop adapter, and future hooks as candidates/hooks. All PRD decisions D1–D11 are DECIDED (D3 and D11 decided PRD v0.6, 2026-10-07; chapter hedges closed in this revision). Rulings recorded 2026-10-11 (Sean): RACK/WACK omitted from the family ACE contract (wrapper auto-pulse deprecated, removal tracked as A5); CD-before-CR convention confirmed.

Scope: The architecture of `amber` (pair-rig top) and `amber_ace` (onyx-rig top) from the CPU GAXI slave through the shared cache core to the fabric masters and ACE snoop responder. The MAS closes the micro-architecture decisions this book defers.

How this book relates to its sources. The amber PRD is the binding requirements document. This HAS carries the Pre-HAS feature sketches (F1–F18) forward, drops their "PROPOSED" hedging where the PRD has decided, and fixes the module set and interfaces as architecture. D3 and D11 are decided (PRD v0.6, 2026-10-07); the chapters record the decided shape and its consequences. The GLOBAL_REQUIREMENTS.md reset-macro, array-syntax, no-bespoke-SRAM, and filelist rules apply throughout.

2.1 Document Organization

Note: All chapters linked below for automated document generation.

2.1.1 Front Matter

- Document Information

2.1.2 Chapter 1: Introduction

- Purpose and Scope
- Document Conventions
- Definitions and Acronyms

2.1.3 Chapter 2: System Overview

- Use Cases
- Key Features
- System Context
- Block Diagram

2.1.4 Chapter 3: Architecture

- Top Structure and the Two Rigs
- Data Flow: Hits, Misses, Fills, and Drains
- Control FSM: Blocking Pipeline and Pending-Fill Bypass
- Coherence FSM and the Snoop Responder
- Arbitration and the Victim Bypass

2.1.5 Chapter 4: Interfaces

- CPU-Side GAXI Slave
- Fabric-Side AXI4 Masters
- Fabric-Side ACE Masters and Coherent Issue
- Snoop Port: AC/CD/CR Adapter
- Observation: MonBus through l1_monlite

2.1.6 Chapter 5: Parameters

- l1_pkg and Build-Time Geometry

2.1.7 Chapter 6: Integration

- Clocking and Reset
- Verification Strategy
- Characterization
- Future Hooks

2.2 Figures

Table 0.0: Figures and their sources

Figure	Source	Subject
2.1	`assets/graphviz/01_block_diagram.dot`	amber shared core, the two rig tops, and the peer/onyx snoop boundary
3.1	`assets/mermaid/02_miss_flow.mmd`	miss-to-fill-to-replay data flow, with victim drain
3.2	`assets/mermaid/03_snoop_flow.mmd`	probe path through tag port B and the pending-

Figure	Source	Subject
		fill bypass

2.3 Related Documentation

Table 0.1: Related documentation

Document	Where	Relationship
amber PRD	../../PRD.md	**binding requirements** — all D-rows and success criteria originate here
amber Pre-HAS	./amber_prehas.md	architecture sketch; this HAS supersedes and ratifies it
amber MAS	../amber_mas/amber_mas_index.md	micro-architecture and pre-RTL signal-contract / kmap workbook
jet PRD	../../../jet-mesi-l1/PRD.md	lockup-free follow-on; amber must not silently pre-decide jet's J-rows
onyx PRD	../../../onyx-ace-ccu/PRD.md	onyx D7 defines the ACE-shaped snoop port; onyx D2 defines the coherent-transaction subset
AMBA ACE Interface Definition	../../../References/AMBA_ACE_Interface_Definition.md	in-repo ACE contract, verified against IHI 0022H
gem5 Ruby protocols	../../../References/gem5-ruby-protocols/README.md	golden-protocol source for the Python reference model and FSM oracles
cache simulator	../../../../../../bin/apps/cache_sim/README.md	golden parity model for LRU/FIFO/RANDOM
Global Requirements	../../../../../../GLOBAL_REQUIREMENTS.md	reset macros, array syntax, no bespoke SRAM, filelist registry

Document	Where	Relationship
amber/jet improvement plan memo	../../../docs/amber-jet-improvement-plan.md	Chinnakonda-informed improvement proposals for amber and jet (2026-10-11); candidates only, nothing decided
Designing a Multiprocessor Cache Subsystem	external: `M. Chinnakonda`	book-informed candidates; sections cited in the chapters that use them

Last Updated: 2026-10-11 **Maintained By:** amber architecture

3 Document Information

This document is the Micro-Architecture Specification (MAS) for `amber`, a parameterised, blocking, snoopy MESI L1 data cache. It closes the implementation decisions that the amber Hardware Architecture Specification (HAS) deliberately deferred to RTL bring-up: pipeline staging, array banking, the exact control FSM state encoding, replacement-policy implementation details, and the pre-RTL combinational decision-logic contracts captured in the kmap workbook.

3.1 References

3.1.1 Related Documents

Related Documents and Specifications

Source	Title	Version
RTL Design Sherpa	amber Product Requirements Document	0.5
RTL Design Sherpa	amber Hardware Architecture Specification	0.7
RTL Design Sherpa	amber Pre-HAS sketch	1.0
ARM	AMBA AXI and ACE Protocol Specification	IHI0022H
gem5	Ruby `MESI_Two_Level` protocol extract	commit f5c5a6e

3.2 Terminology

ACE

AXI Coherency Extensions. The ARM protocol carrying snoop channels (AC/CD/CR) and coherent master transactions.

AC / CD / CR

ACE snoop channels: Address Coherent (AC), Coherent Data (CD), Coherent Response (CR).

CRRESP

5-bit snoop response carrying DataTransfer, PassDirty, IsShared, WasUnique, and an error bit.

DT / PD / IS / WU

CRRESP sub-fields: DataTransfer[0], PassDirty[1], IsShared[2], WasUnique[3].

FUB

Functional Unit Block. A self-contained RTL module with a defined interface and testbench.

GAXI

House generic-AXI streaming pattern (`wr_valid`/`wr_ready`/`wr_data`, `rd_valid`/`rd_ready`/`rd_data`) used for the CPU-side port.

LRU / FIFO / RANDOM / tree-PLRU

Replacement policies supported by `l1_repl`. LRU, FIFO, and RANDOM have cache_sim golden-model parity; tree-PLRU is the timing fallback without a sim golden model today.

MAC

Macro. An integration-level block that instantiates and connects multiple FUBs.

MESI

Cache coherence states: Modified, Exclusive, Shared, Invalid.

MonBus

128-bit packet + 64-bit side-band timestamp monitor bus. `l1_monlite` is a drop-and-count observer that never stalls the observed path.

Pending-fill bypass register

Single register held by `l1_control` while a fill is outstanding, allowing snoops to the same line to be answered with the post-fill state before the array is updated.

RACK / WACK

ACE read/write acknowledge pulses. In amber's adapters they are auto-pulsed one cycle after the last read beat / B handshake. Ruling 2026-10-11 (Sean): the family ACE contract omits RACK/WACK; the auto-pulse is deprecated for family use and its removal is tracked as amber A5.

SDPRAM

Simple dual-port RAM. The house `sdpram_core` primitive is the only allowed tag/data storage.

Snoop responder

`l1_snoop_resp`, the ACE-shaped adapter that presents bus-agnostic `{address, type, response, data}` transactions to `l1_core`.

3.3 Revision History

amber MAS Document Revision History

Rev	Date	Author	Notes
0.5	2026-10-06	seang	Initial amber MAS: index, ch01–ch06, styles, title, and pre-RTL kmap workbook. No RTL yet; all micro-architecture proposals are contracts for the landing implementation.
0.6	2026-10-11	seang	Owner rulings recorded: RACK/WACK omitted from the family ACE contract (auto-pulse deprecated, removal tracked as A5).
0.7	2026-10-15	seang	`l1_*` rename sweep (RTL commit 79dc18ccb; module + file names, chapter files renamed to match).

Rev	Date	Author	Notes
			ACE issue #106 corrections (2594d73bb): ReadNotSharedDirty read misses, PassDirty honoured (install M), CleanUnique/MakeUnique on AR with spec encodings. PF_GUARD documented (BUG-002, ab308d108). Stale dv paths and moved common-ip paths fixed.

Last Updated: 2026-10-11

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4 Architecture Overview

Component: amber — Blocking MESI Snoopy L1 Cache **Version:** 1.0 **Status:** RTL landed; micro-architecture contract for the shared `l1_core` and the amber tops

---

4.1 Introduction

The amber Hardware Architecture Specification (HAS) fixes the module set, interface shapes, and parameter decisions. This Micro-Architecture Specification (MAS) closes what the HAS deferred: the cycle-level behavior of every module, the exact control FSM state encoding, how tag and state pack into the dual-port `sdpram_core`, the replacement-policy datapaths, and the handshake timing of each external interface.

Because amber has **no register block**, there is no CSR programming chapter. All runtime visibility is through the MonBus observation fabric. The MAS therefore replaces the traditional register-map chapter with a MonBus observation map.

4.1.1 What this book covers that the HAS does not

1. **Per-module micro-architecture** — how `l1_control` advances its FSM, how the pending-fill bypass register (`l1_pending_fill_bypass`) is matched and timed, how `l1_tag_array` and `l1_data_array` use `sdpram_core` port A and port B, how `l1_repl` implements each policy, and how `l1_snoop_resp` stages CR and CD.
2. **Cycle-level interface behavior** — GAXI request/response handshake timing, AXI4/ACE master channel timing, AC/CD/CR snoop ordering including the CR-after-CDLAST rule, and MonBus event emission points.
3. **The kmap workbook as decision-logic contract** — every important combinational decision signal is captured as a K-map with axis equations, checked relations, and a mechanically-derived minimal SOP. The workbook is generated by `gen_amber_contracts_kmaps.py`. The workbook was written before the RTL and still carries no `rtl_sop`; now that the RTL has landed, the next step is to diff each derived cover against the RTL expression and apply one of the three methodology verdicts: identical, RTL-redundant, or RTL-differs.

4.2 High-Level Block Diagram

`l1_core` contains all logic shared between the two rig tops. It is the shared L1 core in `cache-ip/common-ip/rtl/top/`, built from the `l1_*` FUBs in `cache-ip/common-ip/rtl/fub/`; amber owns only the tops and `amber_ace_issue`:

```
amber_top  /  amber_ace_top          (amber-mesi-l1/rtl/top/)
+-- l1_core                          (cache-ip/common-ip/rtl/top/)
|   +-- l1_cpu_frontend              (GAXI slave request/response latch)
|   +-- l1_control                   (blocking pipeline FSM)
|   |   +-- l1_pending_fill_bypass   (pending-fill bypass register)
|   |   +-- l1_victim                (depth-1 dirty victim buffer)
|   +-- l1_tag_array                 (dual-port tag + MESI state)
|   +-- l1_data_array                (dual-port data store)
|   +-- l1_repl                      (replacement policy engine)
|   +-- l1_fill                      (line-fill read interface)
|   +-- l1_drain                     (dirty victim write-back interface)
|   +-- l1_snoop_resp                (ACE snoop adapter)
|   +-- l1_monlite                   (MonBus observer)
+-- amber_ace_issue                  (amber_ace_top only; cache events -
> ACE transactions)
+-- axi4_master_rd/wr_monlite        (amber_top)  or
|   axi4ace_master_rd/wr_monlite     (amber_ace_top)
+-- monbus_arbiter                   (merges the l1_monlite, rd and wr
observer streams)
```

The pair-rig top `amber_top` wraps `l1_core` with the house plain-AXI4 read/write masters (`axi4_master_rd_monlite`/`axi4_master_wr_monlite`) toward memory and the snoop responder toward the peer cache. The onyx-rig top `amber_ace_top` wraps the same core with `axi4ace_master_rd_monlite`/`axi4ace_master_wr_monlite` and adds `amber_ace_issue` to map cache-side events to ACE coherent transactions. In both tops a `monbus_arbiter` merges the core's `l1_monlite` stream with the two master-side observer streams onto the single MonBus port.

There is no `RIG` parameter. The rig is selected by instantiating the appropriate top module.

4.3 Key Architectural Concepts

4.3.1 1. Blocking Pipeline

`l1_control` is the only FSM in the cache. A request flows through:

1. **Lookup** — tag/state read on port A.
2. **Hit service** — data read or write on port A, response returned.

3. **Miss resolution** — ask `l1_repl` for a victim way.
4. **Victim drain** — if dirty, stage the victim and launch `l1_drain`.
5. **Fill request** — launch `l1_fill` for a whole-line burst.
6. **Fill completion** — write data and install tag + state on port A.
7. **Replay** — re-present the original request from the front-end latch; it now hits.

Because the pipeline is blocking, no new CPU request is accepted while a miss is in flight. This single-outstanding property is what makes the pending-fill bypass tractable.

4.3.2 2. Dual-Port Arrays

`l1_tag_array` and `l1_data_array` are built from the house `sdpram_core` primitive. Port A serves CPU lookups, fill writes, and replay reads. Port B serves snoop lookups and snoop data readout. The arrays themselves need no arbitration because CPU and snoop use separate ports; contention is resolved at the control FSM level.

4.3.3 3. Snoop Responder and Ordering

`l1_snoop_resp` is the only ACE-aware block inside `l1_core`. It presents a bus-agnostic `{address, type, response, data}` interface to `l1_control`. The adapter completes all CD beats for a transaction before asserting CR, matching the family convention and the cocotb-framework 1.2.0 compliance checker.

4.3.4 4. No Register Block

amber is parameter-only. Configuration, event counting, and debug visibility all come through MonBus. This is deliberate and keeps the CPU-side GAXI port the only software-visible surface.

4.4 Design Decisions

Decision	Source	MAS consequence
GAXI CPU-side slave	PRD D2	`l1_cpu_frontend` is a GAXI slave with a latched request and in-order response stream.
Write-back + write-allocate	PRD D5	Dirty victims are drained; fills are whole-line read-modify-write inside `l1_control`.

Decision	Source	MAS consequence
Plain MESI on 3-bit state	PRD D6	`l1_pkg` encodes M/E/S/I; one bit is reserved for MOESI O-state.
Replacement policies {LRU, tree-PLRU, FIFO, RANDOM}	PRD D7	`l1_repl` implements all four; LRU/FIFO/RANDOM match `cache_sim`; tree-PLRU is the timing fallback.
ACE-shaped snoop port	PRD D4	`l1_snoop_resp` drives AC-ready, CR-valid, CD-valid; CR follows CDLAST.
MonBus observation only	PRD D8	`l1_monlite` emits event packets and drops-and-counts under backpressure.

4.5 Related Documentation

- **amber HAS** — architecture decisions and interface contracts.
- **amber PRD** — binding requirements.
- **K-map workbook** — combinational contracts (written pre-RTL; RTL diff pending).

Last Updated: 2026-10-06

5 Top-Level Port List

This chapter lists the top-level ports of `amber_top` and `amber_ace_top`. Internal `l1_core` ports are defined in the per-block chapters of Chapter 2.

5.1 `amber_top` — Pair-Rig Top

5.1.1 Clock and Reset

Signal	Direction	Width	Description
`aclk`	input	1	System clock
`aresetn`	input	1	Active-low asynchronous reset, synchronous deassert

5.1.2 CPU GAXI Slave

Signal	Direction	Width	Description
`cpu_req_wr_valid`	input	1	CPU request valid
`cpu_req_wr_ready`	output	1	Cache ready to accept request
`cpu_req_wr_data`	input	`CPU_REQ_W`	Packed {addr, we, be, wdata}
`cpu_rsp_rd_valid`	output	1	Cache response valid
`cpu_rsp_rd_ready`	input	1	CPU ready to accept response
`cpu_rsp_rd_data`	output	`CPU_RSP_W`	Packed {rdata}

5.1.3 AXI4 Memory Masters

Plain AXI4 masters driven by `l1_fill` and `l1_drain` through the house `axi4_master_rd_monlite` / `axi4_master_wr_monlite` wrappers, whose signal names and widths they follow.

Channel	Direction	Signals
AR	out	`m_axi_arid`,

Channel	Direction	Signals
		m_axi_araddr, m_axi_arlen, m_axi_arsize, m_axi_arburst, m_axi_arlock, m_axi_arcache, m_axi_arprot, m_axi_arqos, m_axi_arregion, m_axi_aruser, m_axi_arvalid
R	in/out	m_axi_rid, m_axi_rdata, m_axi_rresp, m_axi_rlast, m_axi_ruser, m_axi_rvalid, m_axi_rready
AW	out	m_axi_awid, m_axi_awaddr, m_axi_awlen, m_axi_awsize, m_axi_awburst, m_axi_awlock, m_axi_awcache, m_axi_awprot, m_axi_awqos, m_axi_awregion, m_axi_awuser, m_axi_awvalid
W	out	m_axi_wdata, m_axi_wstrb, m_axi_wlast, m_axi_wuser, m_axi_wvalid
B	in/out	m_axi_bid,

Channel	Direction	Signals
		m_axi_bresp, m_axi_buser, m_axi_bvalid, m_axi_bready

5.1.4 ACE Snoop Responder

Signal	Direction	Width	Description
m_axi_acaddr	input	ADDR_WIDTH	Snoop address
m_axi_acsnoop	input	4	Snoop transaction type
m_axi_acprot	input	3	Snoop protection
m_axi_acvalid	input	1	Snoop address valid
m_axi_acready	output	1	Cache ready for snoop
m_axi_crresp	output	5	Snoop response
m_axi_crvalid	output	1	Snoop response valid
m_axi_crready	input	1	Snoop response ready
m_axi_cddata	output	DATA_WIDTH	Snoop data
m_axi_cdlast	output	1	Snoop data last beat
m_axi_cdvalid	output	1	Snoop data valid
m_axi_cdready	input	1	Snoop data ready

5.1.5 MonBus

Signal	Direction	Width	Description

mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor consumer ready
mon_packet	output	128	Monitor packet
mon_timestamp	output	64	Side-band timestamp

5.2 amber_ace_top — Onyx-Rig Top

amber_ace_top has the same CPU GAXI slave, snoop responder, and MonBus ports as amber_top. The memory-side masters are full ACE masters with ARSNOOP / AWSNOOP, ARDOMAIN / AWDOMAIN and ARBAR / AWBAR fields, the 4-bit ACE RRESP (IHI 0022H §D2.1), and auto-pulsed RACK / WACK outputs (deprecated per ruling 2026-10-11 (Sean); removal tracked as amber A5). amber_ace_issue maps cache-side events to ACE transactions; see Chapter 2.8.

Signal	Direction	Width	Description
m_axi_arsnoop	output	4	ACE read transaction type
m_axi_ardomain	output	2	Shareability domain; tied to 0b10 Outer Shareable
m_axi_arbar	output	2	Barrier type; tied to 0b00, normal access (amber issues no barriers)
m_axi_rresp	input	4	[1:0] AXI response, [2] PassDirty, [3] IsShared; the core consumes [1:0] only (Chapter 3.3)
m_axi_rack	output	1	ACE read acknowledge (auto-pulsed)
m_axi_awsnoop	output	3	ACE write transaction type
m_axi_awdomain	output	2	Shareability domain; tied to 0b10 Outer Shareable

Signal	Direction	Width	Description
`m_axi_awbar`	output	2	Barrier type; tied to `0b00`, normal access
`m_axi_wack`	output	1	ACE write acknowledge (auto-pulsed)

All other AXI/ACE master signals mirror `amber_top` but connect to `axi4ace_master_rd_monlite` / `axi4ace_master_wr_monlite` instead of the plain AXI4 wrappers.

5.3 Parameters

All geometry, policy, and bus-width choices are elaboration parameters. See Chapter 6 and the HAS parameters chapter for the complete list. There is no runtime register block.

Last Updated: 2026-10-11

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6 Clocks and Reset

6.1 Clock Domains

Table 1.3.1: Clock domains

Clock	Frequency	Usage
`aclk`	50–200 MHz	Primary — all amber logic, AXI4/ACE/GAXI interfaces, MonBus

amber is a single-clock IP. The GAXI slave, AXI4/ACE masters, snoop responder, and MonBus all run on `aclk`.

6.2 Reset

Table 1.3.2: Reset signals

Signal	Polarity	Type	Usage
`aresetn`	Active-low	Async assert, **synchronized internally**	Primary reset input for all amber logic and memories

`aresetn` is an asynchronous-assert input whose deassertion quality is not trusted: `amber_top` runs it through the house `reset_sync` (`rtl/cdc`, N=3, `ASYNC_REG`) before any internal block sees it — as does `amber_ace_top`, the ACE-rig twin — so integrators may deassert it at any phase — including asynchronously or mid-cycle — and the release the core sees is synchronous. Assertion remains asynchronous end to end. The Genesys2 pair-rig top carries its own synchronizer upstream; cascading is harmless (adds N cycles of release latency).

`aresetn` resets all flops. The `sdpram_core` instances inside `l1_tag_array` and `l1_data_array` are not reset; their contents are invalidated by clearing valid bits or by the first access after reset, depending on the implementation choice recorded in Chapter 2.3.

All sequential logic uses the house `ALWAYS_FF_RST` macro with `RST_ASSERTED` on the synchronized internal reset. This is the same pattern used in the AMBA wrappers and STREAM.

The internal synchronization is verified by the directed test `dv/tests/test_amber_top_reset_sync.py`: an asynchronous (mid-cycle) deassert must delay the post-reset init walk by exactly N cycles, and a mid-cycle assertion must re-enter the init state without waiting for a clock edge.

Last Updated: 2026-10-14

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7 l1_control: Blocking Pipeline FSM

Module: `l1_control.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/`
Status: Landed (Task 3: CPU path; Task 4: snoop-during-fill service + pending-fill bypass leaf); scored against the Task 2 gem5-derived oracle at gate/func/full, geometries s16w2 + s128w4

7.1 Overview

`l1_control` is the cache's only state machine. It owns the blocking pipeline, the tag-array port-A schedule, the pending-fill bypass register, victim-way selection, and the handshakes to `l1_fill`, `l1_drain`, `l1_victim`, and `l1_snoop_resp`. Because the cache is blocking, there is exactly one CPU transaction in flight at any time; snoops are serviced concurrently on tag-array port B.

7.2 State Encoding

`l1_pkg` defines a 3-bit cache-line state field for `M`, `E`, `S`, `I`. One extra code is reserved for a future MOESI `O` state; the control FSM v1.0 implements only MESI.

```
typedef enum logic [2:0] {
    STATE_I = 3'b000,  // Invalid
    STATE_S = 3'b001,  // Shared
    STATE_E = 3'b010,  // Exclusive
    STATE_M = 3'b011,  // Modified
    STATE_O = 3'b100,  // Reserved for future MOESI Owned state
    // 3'b101, 3'b110, 3'b111 reserved
} cache_state_t;
```

The 3-bit width is intentional: it gives one reserved code for `O` and three spare codes for future transient/stable elaborations without widening the tag store.

7.2.1 FSM states inside l1_control

The FSM itself is one-hot encoded. Stable states are the ones that can persist between CPU requests; transient states exist only while a miss or snoop is in progress, mirroring the gem5 Ruby `MESI_Two_Level-L1cache.sm` philosophy that every in-flight condition is an explicit enumerated state.

State	Type	Meaning
`CTRL_IDLE`	stable	Waiting for a CPU request from `l1_cpu_frontend`.
`CTRL_LOOKUP`	transient	Tag/state lookup on port A;

State	Type	Meaning
		hit/miss decision.
CTRL_HIT_RD	transient	Read hit: fetch data from l1_data_array port A, return response.
CTRL_HIT_WR	transient	Write hit: update data, promote to M, return response.
CTRL_MISS_VICTIM	transient	Miss: ask l1_repl for victim way; read victim state.
CTRL_MISS_DRAIN	transient	Victim is dirty: stage it in l1_victim, launch l1_drain.
CTRL_MISS_FILL	transient	Launch l1_fill, wait for RLAST.
CTRL_FILL_WRITE	transient	Write fill data into l1_data_array port A; install tag + state.
CTRL_REPLAY	transient	Re-present original request from front-end latch; it now hits.
CTRL_SNOOP	transient	Service a snoop on port B (can preempt / interleave with miss states).
CTRL_ERROR	stable	Sticky fatal error; cleared only by reset.

The FSM is not a minimal two-state controller; the explicit transient states are the formal proof surface. Each state asserts exactly the control signals that are legal for that state, and illegal states default to CTRL_ERROR.

7.3 Pipeline Staging

The HAS deferred pipeline staging to the MAS. This section fixes it.

7.3.1 Stage 0: Request latch

`l1_cpu_frontend` accepts a GAXI request and holds `{addr, we, be, wdata}` in a register. `l1_control` sees a single `req_valid` pulse and the latched request fields. The front end does not accept a new request until `req_ready` is returned.

7.3.2 Stage 1: Tag lookup (CTRL_LOOKUP)

On the cycle after request acceptance, `l1_control` drives `ctrl_tag_a_set`; the tag array returns per-way `{tag, state}` combinationally on `ctrl_tag_a_tag_state`. The lookup also selects the victim way from `l1_repl` combinatorially so that the victim is known on the same cycle the hit/miss decision is made.

7.3.3 Stage 2: Hit or miss decision

- **Hit:** transition to `CTRL_HIT_RD` or `CTRL_HIT_WR`. The data array is accessed in the same cycle as the state check (the set/way was already decoded from the lookup).
- **Miss:** transition to `CTRL_MISS_VICTIM`.

7.3.4 Stage 3: Miss handling

1. `CTRL_MISS_VICTIM`: read the victim tag/state from the tag array and, if dirty, copy the victim line data from `l1_data_array` port A into `l1_victim`.
2. `CTRL_MISS_DRAIN`: if the victim is M, launch `l1_drain` with the address and data from `l1_victim`. Wait for the drain-done handshake.
3. `CTRL_MISS_FILL`: launch `l1_fill` with the missing line address. The pending-fill bypass register is loaded now.
4. `CTRL_FILL_WRITE`: on `fill_done`, write the fill data into `l1_data_array` port A and install the tag + new state in `l1_tag_array` port A.
5. `CTRL_REPLAY`: re-assert `req_valid` to the hit path using the original latched request; the next cycle is `CTRL_LOOKUP` again and the line is present.

7.3.5 Snoop interleaving

`l1_control` resolves CPU-vs-snoop contention with snoop priority. A pending snoop can cause the CPU pipeline to stall for one cycle at safe boundaries (after a tag-array access completes, never mid-burst). The snoop path uses `tag_b_*` and `data_b_*` ports exclusively; it never needs port A except when a snoop triggers an invalidation or downgrade, which is applied on the next available port-A cycle.

Landed behavior (Task 4): the grant is restricted to safe waiting states — `CTRL_IDLE`, `CTRL_MISS_FILL` (once the bypass is armed, or while an upgrade waits on its acknowledge), and `CTRL_MISS_DRAIN` — so the pipeline stalls exactly one cycle at the boundary plus the CD transfer

time. The response reference state resolves as: pending-fill bypass match (answered at the post-fill `pf_state`, CD beats sourced from the fill way and gated per-beat by `pf_data_valid`, stalled until the beat arrives) > the stale victim entry during a fill (no transfer — the writeback already completed, gem5 `M_I × WB_Ack → I`) > the port-B tag hit (installed state; a dirty victim still draining is served at M from the staged copy, `SINK_WB_ACK`). An invalidating/downgrading snoop on the pending line is remembered (`pend_vld`/`pend_state`, invalidation-sticks) and applied at fill commit; on an installed line it is written on port A inside the service. CR presentation after CDLAST is `l1_snoop_resp`'s contract, not control's.

7.4 Key Control Signals

Landed port names (the table is checked against `l1_control.sv`). Read data (tag/state, hit read data) is combinational on the landed arrays, so the port-A schedule is address + write strobe, no read enable.

Signal	Direction	Meaning
`ctrl_tag_a_set`	to `l1_tag_array`	Port A lookup set index.
`ctrl_tag_a_tag_state`	from `l1_tag_array`	Per-way `{tag, state}` lookup result.
`ctrl_tag_a_wr_en`	to `l1_tag_array`	Port A write enable (init walk / install / promotion).
`ctrl_tag_a_wr_tag_state`	to `l1_tag_array`	Port A write data {tag, state}.
`ctrl_tag_a_wr_way_onehot`	to `l1_tag_array`	Way select for write (all-ones on the init walk).
`ctrl_data_a_addr`	to `l1_data_array`	Port A address {set, beat}.
`ctrl_data_a_way`	to `l1_data_array`	Port A read way (hit way or victim way).
`ctrl_data_a_rdata`	from `l1_data_array`	Hit read data (combinational).
`ctrl_data_a_wr_en`	to `l1_data_array`	CPU write-merge enable (hit / replayed D-4 merge).

Signal	Direction	Meaning
ctrl_data_a_wr_wdata	to l1_data_array	Merge write data.
ctrl_data_a_wr_be	to l1_data_array	Per-byte merge enables.
ctrl_repl_req	to l1_repl	Request replacement decision (one-cycle pulse per miss).
ctrl_repl_way	from l1_repl	Selected victim way.
ctrl_repl_hit / ctrl_repl_update	to l1_repl	Policy update: hit service / fill install (upgrade fires hit).
ctrl_victim_load	to l1_victim	Strobe: stage the dirty victim line from the data array.
ctrl_fill_start	to l1_fill	Launch a fill for this line address.
ctrl_fill_addr	to l1_fill	Line-aligned fill address.
ctrl_req_class	to l1_fill / amber_ace_issue	Request class of the in-flight miss (READ_SHARED / READ_UNIQUE / CLEAN_UNIQUE).
ctrl_fill_done	from l1_fill	Fill completed (RLAST accepted).
ctrl_fill_beat_valid / ctrl_fill_beat_idx	from l1_fill	Received-beat strobe + index; feeds the bypass pf_data_valid (MAS ch02/06).
ctrl_drain_start	to l1_drain	Launch a drain from l1_victim.
ctrl_drain_done	from l1_drain	Drain completed (B handshake).
ctrl_tag_b_req	to l1_tag_array (via core mux)	Port-B ownership: high in the snoop grant cycle

Signal	Direction	Meaning
		and throughout CTRL_SNOOP.
ctrl_tag_b_set	to l1_tag_array	Port-B lookup set (snoop set).
ctrl_tag_b_tag_state	from l1_tag_array	Per-way {tag, state} snoop lookup result (combinational).
ctrl_data_b_addr / ctrl_data_b_way	to l1_data_array	Port-B CD read {set, beat} at the hit way (or the fill way for a bypass hit).
ctrl_data_b_rdata	from l1_data_array	CD beat data (combinational).
ctrl_snoop_req	from l1_snoop_resp	Snoop address + type valid; held until ctrl_snoop_ready.
ctrl_snoop_ready	to l1_snoop_resp	Control accepts the snoop this cycle (safe-boundary grant).
ctrl_crresp	to l1_snoop_resp	5-bit CRRESP result, combinational at the grant cycle (latched by the adapter).
ctrl_cddata / ctrl_cdlast / ctrl_cdvalid	to l1_snoop_resp	CD beat data/last/valid (each beat gated servable; CR is presented by the adapter after CDLAST).
ctrl_cdready	from l1_snoop_resp	CD beat accepted.
ctrl_req_ready	to l1_cpu_frontend	Ready for the next CPU request (IDLE, init done).
ctrl_rsp_valid	to l1_cpu_frontend	Response valid (one-

Signal	Direction	Meaning
		cycle pulse).
`ctrl_rsp_data`	to `l1_cpu_frontend`	Response read data / write echo.
`ctrl_init_busy` / `ctrl_init_set`	status	Init-walk in progress / current set.
`ctrl_state`	status	FSM state (pkg `ctrl_state_t` encoding, observability).

7.5 Miss-Path Decision K-map

The combinational decision that launches drain/fill/replay is captured in the kmap workbook:

- **Outputs:** `start_drain`, `start_fill`, `replay_now`.
- **Axes:** `{hit, victim_dirty, pending_bypass_match}`.
- **Sheet:** `K-maps amber control` in `amber_signal_contracts.xlsx`, generated by `gen_amber_contracts_kmaps.py`.

The sheet is marked **PROPOSED** because it is a micro-architecture decision owned by this chapter. The RTL (`l1_control`) has landed, but the workbook does not yet carry its `rtl_sop`, so the derived cover has not been diffed against it (Chapter 5).

Last Updated: 2026-10-07 (Task 4: snoop service + bypass integration)

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8 Pending-Fill Bypass Register

Module: `l1_pending_fill_bypass.sv` (leaf, instantiated by `l1_control`) **Location:** `projects/components/cache-ip/common-ip/rtl/fub/` **Status:** RTL landed (Task 4); built as a leaf (DECISION D-2) so Task 13 proves it standalone. Integration scenarios: `dv/testplans/amber_pending_fill_bypass_testplan.yaml` (via the `l1_control` suite).

8.1 Purpose

While a fill is outstanding, a snoop may arrive for the same line. The pending-fill bypass register lets `l1_control` answer that snoop with the post-fill state and any fill beats already received, before the tag and data arrays have been updated. This is the only place in amber where a line can be observed in a state that has not yet been committed to the arrays, so its correctness is a SymbiYosys target (Task 13).

8.2 Register Fields

Field	Width	Meaning
`pf_addr`	`ADDR_WIDTH - OFFSET_WIDTH`	Line address (tag + set index) of the fill.
`pf_state`	3	State that will be installed when the fill completes.
`pf_data_valid`	`FILL_BEATS`	Per-beat mask: bit[i] is 1 when fill beat i has been received.
`pf_active`	1	Register is valid and the fill is still outstanding.

The register is loaded in `CTRL_MISS_FILL` when the fill is launched. `pf_state` is determined by the request type:

- Read miss to shared → `STATE_S`
- Read miss to exclusive → `STATE_E` (arrives with the Task 5 fill FUB's exclusive qualifier; unreachable in the Task 4 integration, which installs S deterministically)
- Write miss (whole-line allocate) → `STATE_M`

An upgrade (`CLEAN_UNIQUE`) does NOT load the register (MINOR-1 decision, pinned by the `UpgradeNoBypassArm` directed test): an upgrade carries no fill data in flight — the line stays installed in S and snoops are answered from the tag array on port B — and a bypass armed with no beats would deadlock the CD channel against `pf_data_valid` bits that can never set. The load is qualified `!upgr_q` in `l1_control`.

`pf_data_valid` is updated as each R beat is accepted by `l1_fill` (the `ctrl_fill_beat_valid`/`ctrl_fill_beat_idx` strobe, MAS ch02/06). The snoop responder can forward fill beats whose `pf_data_valid` bit is set; beats not yet received stall the CD channel until they arrive.

8.3 Match Logic

A snoop matches the pending-fill bypass when:

```
pf_active &&
(snoop_line_addr == pf_addr)
```

The line address is `{tag, set_index}`; the line offset is ignored. The kmap workbook contains a small optional bypass-match sheet, but the match is a single equality compare; the sheet is included only if it earns its place as a teaching aid.

8.3.1 PF_GUARD: grant against the fill only when the fill is real

`l1_control` grants a matching snoop against the pending fill only when the fill is actually in flight (`PF_GUARD=1`, default `1'b0`; parameter on both `l1_control` and `l1_core`; `amber_top` sets `1'b1`). A fill whose AR is still seeking acceptance can never produce beats while a fabric parks in its response state waiting for this very snoop — promising data from it deadlocked the pair rig (amber BUG-002, commit `ab308d108`). With the guard enabled, a pre-acceptance match is answered from the installed tag state instead, and the fabric serializes the two fills. Standalone benches without such a fabric leave `PF_GUARD=0` and keep the classic behavior.

8.4 Timing

Cycle	Event
`CTRL_MISS_FILL` entry	`pf_active` set, `pf_addr` and `pf_state` loaded, `pf_data_valid` cleared.
Each accepted R beat	Corresponding `pf_data_valid` bit set by `l1_fill` done-strobe.
Snoop arrives during fill	If match, `CRRESP` and CD beats sourced

Cycle	Event
	from bypass register; otherwise normal tag lookup.
CTRL_FILL_WRITE	Bypass register contents written to arrays; pf_active cleared.
CTRL_REPLAY	Original request now hits the installed line.

The critical correctness property is **state-accuracy**: the bypass register must answer with the post-fill state, never the pre-fill (Invalid) state. A snoop that would invalidate the line after the fill must see the post-fill state and then be allowed to perform its invalidation in the normal snoop path. The invalidation/downgrade itself is NOT a register field (MAS ch02/02 names exactly the four fields above): it is control-side context (pend_vld/pend_state), armed by a snoop whose Table-3.0 next-state differs from the resolved reference state and consumed at CTRL_FILL_WRITE — **invalidation-sticks**: once armed, a later downgrade snoop does not overwrite an armed I (gem5 IS_I .sm:1390; mapping-notes divergences 3/8; the _im_step fixed-point corner is pinned by the ImStepPendingClearCorner directed test and five new oracle table rows).

8.5 Snoop vs Fill Ordering

If a snoop arrives for the pending line and requires data transfer, the CD beats are sourced from whichever copy is available:

- If the beat has already been received (pf_data_valid[i] == 1), it is forwarded directly from the fill-path register.
- If the beat has not yet arrived, the snoop responder stalls on cd_ready until the beat arrives, then forwards it.

The fill still completes normally into the arrays after RLAST. The snoop does not cancel the fill; any state change caused by the snoop is applied after the fill commits.

Last Updated: 2026-10-07 (Task 4: RTL landed)

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9 l1_tag_array and l1_data_array

Modules: `l1_tag_array.sv`, `l1_data_array.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/` **Status:** RTL landed 2026-10-06; this chapter is the micro-architecture contract the RTL implements. DV: `dv/tests/test_amber_tag_array.py` + `test_amber_data_array.py`, green at gate/func/full across the default and tiny-formal geometries.

9.1 Overview

Both arrays are per-way inferred RAMs — the same storage idiom the house FIFOs and `sdpram_core` use internally, with the GLOBAL_REQUIREMENTS 1.2 synthesis attributes. They are **not** `sdpram_core` instances: `sdpram_core` (`rtl/amba/shared`) is a FUB/AXI burst *slave*, the wrong shape for a multi-way parallel tag lookup or single-beat random data readout. Each array has one synchronous write port (one-hot way select) and two combinational lookup ports: port A serves CPU/fill traffic and port B serves snoop traffic, matching the port assignment below. Combinational read keeps the hit/miss decision and the snoop CRRESP path inside one cycle; it maps to distributed RAM (tags) / block RAM (data) rather than a wrapped burst slave.

The one-write / two-read shape exceeds a single physical 2-port BRAM; per-way flat storage (`way * SETS + set` for tags, `way * (SETS*FILL_BEATS) + {set, beat}` for data) lets synthesis bank per way at the default geometry. That is the HAS F8 dual-port decision implemented at L1-array granularity.

9.2 l1_tag_array

9.2.1 Tag + state packing

Each way has its own `sdpram_core` instance, or the ways are banked into one instance per way depending on the synthesis attributes allowed by `GLOBAL_REQUIREMENTS.md`. The data width of each instance is:

```
TAG_STATE_WIDTH = TAG_WIDTH + 3
```

where `TAG_WIDTH = ADDR_WIDTH - SET_INDEX_WIDTH - LINE_OFFSET_WIDTH` and the 3 bits carry `cache_state_t`.

The memory word is packed as `{tag[TAG_WIDTH-1:0], state[2:0]}` — verified by the DV against per-location values that place nonzero patterns in both fields.

9.2.2 Port assignment

Port	Traffic	Operations
A	CPU / fill	Read tag+state for lookup; write tag+state on fill or hit-state update.
B	Snoop	Read tag+state for snoop lookup.

Port A is driven by `l1_control`. Port B is driven by `l1_control` during snoop handling.

9.2.3 Addressing

The address into the tag array is the set index:

```
tag_addr = req_addr[SET_INDEX_WIDTH + OFFSET_WIDTH - 1 : OFFSET_WIDTH]
```

Way selection is done by instantiating one `sdpram_core` per way and decoding the way index on writes. On reads, all ways return their tag+state in parallel and a comparator determines the hit way.

9.2.4 Banking

For timing closure at larger geometries, each way is a separate physical memory instance. There is no cross-way banking; a fill or snoop reads all ways in parallel and writes one. This keeps the critical hit/miss decision a simple combinational comparison across `WAYS` tag outputs.

9.3 l1_data_array

9.3.1 Organization

The data array stores cache-line data. A line is `LINE_BYTES*8` bits wide and is split into `FILL_BEATS = LINE_BYTES / (BUS_WIDTH/8)` beats. Each way has its own `sdpram_core` instance, or one instance per way, with:

```
DATA_MEM_DEPTH = SETS * FILL_BEATS
DATA_MEM_WIDTH = BUS_WIDTH
```

The address is `{set_index, beat_index}`.

9.3.2 Port assignment

Port	Traffic	Operations
A	CPU / fill	Read data on hit or replay; write fill

Port	Traffic	Operations
		beats.
B	Snoop	Read whole line for snoop data transfer.

9.3.3 Write masking

`sdpram_core` is instantiated with byte-write enable support. On a CPU write hit, only the bytes selected by `be` are written; the rest of the line is preserved. On a fill, the entire beat is written with all byte enables asserted. On a write-allocate miss, the fill data is merged with the CPU write data before installation.

9.3.4 Snoop data readout

A snoop that requires data transfer reads the matching way from port B beat-by-beat. `l1_control` drives `b_addr = {snoop_set, beat}`, increments the beat counter each cycle until the line is complete, and supplies the hit way on `b_way` (from the tag compare on port B). The data is forwarded to `l1_snoop_resp` for CD transmission.

9.4 Reset Behavior

The arrays carry **no reset port** (GLOBAL_REQUIREMENTS 1.4 — SRAM contents are not reset, and real BRAM has no reset pin). After `aresetn` deassertion, `l1_control` treats all ways as Invalid until the init walk completes; option 1 from the pre-RTL review is the contract: a hardware sequence walks all sets and writes `STATE_I` to every way (the DV TBs perform the same walk so unwritten locations cannot leak X's). The formal proofs assume all lines are Invalid after reset.

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10 l1_repl: Replacement Policy Engine

Module: `l1_repl.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/`
Status: RTL landed 2026-10-06; this chapter is the micro-architecture contract the RTL implements. DV: `dv/tests/test_amber_repl.py`, green at gate/func/full across all four policies x {128/4, 16/2, 64/8} geometries. The TB carries an independent Python golden model per policy (tree-PLRU included — it still has no cache_sim golden; the TB model is the check).

10.1 Overview

`l1_repl` selects the victim way on a cache miss. The policy is an elaboration parameter `REPL_POLICY` with supported values `lru`, `tree_plru`, `fifo`, and `random`. LRU is the default; tree-PLRU is the timing-tight fallback.

In the RTL the parameter is declared `int` holding the `amber_repl_t` encoding from `l1_pkg` (house precedent: `fifo_sync` MEM_STYLE is typed, but a typed enum parameter receives `-G` overrides as raw 32-bit constants and verilator WIDTHTRUNCs values 2/3 — the int declaration keeps test-time overrides width-clean; the generate branches compare against `int'(AMBER_REPL_*)` casts).

10.2 Interface

Signal	Direction	Width	Meaning
`repl_req`	input	1	Request a victim way this cycle.
`repl_set`	input	`SET_INDEX_WIDTH`	Set index for the access.
`repl_hit_way`	input	`WAY_WIDTH`	Way that hit (for update).
`repl_hit`	input	1	A hit occurred; update policy state.
`repl_victim_way`	output	`WAY_WIDTH`	Selected victim way.
`repl_update`	input	1	A miss install occurred; update

Signal	Direction	Width	Meaning
			policy state for `repl_set`.

`repl_hit` and `repl_update` are mutually exclusive in the blocking pipeline: a cycle is either a hit service or a miss install, never both.

10.3 LRU (True LRU)

True LRU maintains a recency stack per set. For `WAYS` ways, each set stores `WAYS*(WAYS-1)/2` ordering bits. On every hit or install, the accessed way is moved to the most-recent position and the other ways are shifted down. The victim is the least-recent way.

This matches `cache_sim` exactly and is the golden reference for policy parity.

10.3.1 Implementation

Use a per-set array of `WAYS` rank values. The rank value is `0` for most-recent, `WAYS-1` for least-recent. On access:

```
for (int w = 0; w < WAYS; w++) begin
    if (w == accessed_way)
        rank[set][w] = 0;
    else if (rank[set][w] < old_rank_of_accessed_way)
        rank[set][w] += 1;
end
```

The victim way is the one with `rank[set][w] == WAYS-1`.

10.4 FIFO

Each set maintains a FIFO queue of way indices, stored as a ring with a head pointer. On a miss, the head of the queue is the victim; on an install, the ring slot at the head is overwritten with the installed way and the head advances — the popped victim slot becomes the tail, so one write plus one head bump is the whole update. On a hit, the queue is unchanged (FIFO does not recency-stack); the TB golden model encodes the same rule (`UPDATES_ON_HIT = False`).

This matches `cache_sim` exactly, under the invariant that the installed way equals the victim way — which `l1_control` guarantees because fills install into the victim way. Installing into any other way would require a mid-queue removal this ring does not perform; that is a contract, not an accident.

This matches `cache_sim` exactly.

10.5 RANDOM

Each set contains a small LFSR. On a replacement request, the LFSR value is truncated to `WAY_WIDTH` bits to select the victim way. The LFSR advances every replacement request. The seed is parameterizable for repeatable tests.

This matches `cache_sim` exactly when the same LFSR polynomial and seed are used.

10.6 Tree-PLRU

Tree-PLRU is carried as the timing-tight fallback per PRD D7. It uses `WAYS-1` bits per set arranged as a binary tree. On each access, the tree bits point away from the accessed way. On a replacement request, the tree bits are followed to find the victim.

Tree-PLRU does **not** have a `cache_sim` golden model today. It is cross-checked by RTL self-checks and is recorded as a future `cache_sim` extension. This is an honest caveat, not a hidden assumption.

10.7 Policy Selection

The policy is selected at elaboration time; the RTL declares the parameter `int` (values are the `amber_repl_t` encodings) and generates exactly one branch:

```
parameter int REPL_POLICY = int'(AMBER_REPL_LRU)   // values:
amber_repl_t

generate
    if (REPL_POLICY == int'(AMBER_REPL_LRU)) begin : gen_lru
        // true LRU rank array
    end else if (REPL_POLICY == int'(AMBER_REPL_FIFO)) begin :
gen_fifo
        // FIFO ring array
    end else if (REPL_POLICY == int'(AMBER_REPL_RANDOM)) begin :
gen_random
        // LFSR per set
    end else if (REPL_POLICY == int'(AMBER_REPL_TREE_PLRU)) begin :
gen_tree_plru
        // binary tree PLRU
```

```
    end
endgenerate
```

Only one policy is instantiated per build. This keeps area predictable and matches the `cache_sim` per-policy parity model.

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11 l1_victim: Depth-1 Victim Buffer

Module: `l1_victim.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/`
Status: RTL landed; this chapter is the micro-architecture contract the RTL implements.

11.1 Overview

`l1_victim` is a depth-1 buffer that holds one dirty cache line while its write-back (drain) is outstanding. The depth-1 size matches the single-outstanding-miss model: there can be only one dirty victim at a time because the pipeline blocks on the miss until it resolves.

11.2 Fields

Field	Width	Meaning
`victim_addr`	`ADDR_WIDTH`	Full line address (for AW).
`victim_data`	`LINE_BYTES*8`	Dirty line data.
`victim_valid`	1	Buffer is occupied.

11.3 Handshake with l1_control

Signal	Direction	Meaning
`victim_load`	input from `l1_control`	Load the buffer from `l1_data_array` port A this cycle.
`victim_addr_in`	input from `l1_control`	Line address to load.
`victim_data_in`	input from `l1_control`	Line data to load.
`victim_busy`	output to `l1_control`	Buffer already occupied.
`victim_empty`	output to `l1_control`	Buffer is empty.

`victim_load` asserts for one cycle in `CTRL_MISS_VICTIM` when the victim is dirty. The buffer is busy until `l1_drain` reports done.

11.4 Bypass for Snoops

If a snoop arrives for the victim line while the drain is still outstanding, `l1_victim` can supply the data without re-reading `l1_data_array`. `l1_control` checks:

```
victim_valid && (snoop_line_addr == victim_addr)
```

If matched, the snoop responder sources CD beats from `victim_data` instead of from `data_array` port B. This is the victim bypass.

11.5 Timing

Cycle	Event
`CTRL_MISS_VICTIM`, victim is M	`victim_load` asserted; `victim_valid` sets next cycle.
`CTRL_MISS_DRAIN`	`l1_drain` consumes `victim_addr` and `victim_data`; begins AW/W burst.
Drain done (`ctrl_drain_done`)	`victim_valid` cleared.

The buffer is never loaded while valid; the single-outstanding property guarantees this cannot happen.

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12 l1_fill and l1_drain

Modules: `l1_fill.sv`, `l1_drain.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/` **Status:** RTL landed; this chapter is the micro-architecture contract the RTL implements.

12.1 l1_fill

12.1.1 Purpose

`l1_fill` issues whole-line read bursts to the memory-side master and forwards received beats to `l1_data_array` port A and to the pending-fill bypass register.

12.1.2 Interface

Signal	Direction	Meaning
`fill_start`	input from `l1_control`	Launch a fill for `fill_addr`.
`fill_addr`	input from `l1_control`	Line-aligned address.
`fill_done`	output to `l1_control`	All beats received (RLAST accepted).
`fill_beat_valid`	output to `l1_control` / data array	A fill beat is valid this cycle.
`fill_beat_data`	output	Fill beat data.
`fill_beat_idx`	output	Beat index within line.
`fill_last`	output	Last beat of burst.

12.1.3 Burst parameters

- `ARLEN = FILL_BEATS - 1`
- `ARSIZE = log2(BUS_WIDTH/8)`
- `ARBURST = INCR`
- `ARADDR` line-aligned

12.1.4 Timing

1. `fill_start` asserted in `CTRL_MISS_FILL`.
2. `l1_fill` asserts `fub_axi_arvalid` to the memory-side read master.
3. As R beats arrive, `fill_beat_valid` pulses; `fill_beat_idx` increments.

4. Each beat is written to `l1_data_array` port A and the `pf_data_valid` mask is updated.
5. On `RLAST` acceptance, `fill_done` pulses for one cycle.

12.2 l1_drain

12.2.1 Purpose

`l1_drain` takes a dirty line from `l1_victim` and issues a whole-line write-back burst to the memory-side write master.

12.2.2 Interface

Signal	Direction	Meaning
`drain_start`	input from `l1_control`	Launch drain from `l1_victim`.
`drain_done`	output to `l1_control`	B response received.
`drain_beat_ready`	input from memory master	W-channel ready.
`drain_beat_valid`	output	W-channel valid.
`drain_beat_data`	output	W-channel data.
`drain_beat_strb`	output	W-channel strobe (all-1s for whole-line).
`drain_beat_last`	output	W-channel last.

12.2.3 Burst parameters

- `AWLEN = FILL_BEATS - 1`
- `AWSIZE = log2(BUS_WIDTH/8)`
- `AWBURST = INCR`
- `AWADDR = victim_addr`
- `WSTRB = all-1s` for every beat

12.2.4 Timing

1. `drain_start` asserted in `CTRL_MISS_DRAIN` when `l1_victim` is valid.
2. `l1_drain` asserts `fub_axi_awvalid` and begins driving W beats as the master accepts them.
3. After `WLAST` is accepted and the B response arrives, `drain_done` pulses.

12.2.5 On the onyx rig

For `amber_ace_top`, dirty evictions are issued as `WriteBack` transactions. `amber_ace_issue` supplies `AWSNOOP` and the AW address; `l1_drain` supplies the W beats. Clean evictions are `Evict` transactions with no data.

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13 l1_snoop_resp: Snoop Responder Adapter

Module: `l1_snoop_resp.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/` **Status:** RTL landed 2026-10-06; this chapter is the contract the RTL implements. Wraps the house `axi4ace_snoop_slave` transport (which exists in `rtl/amba/ace` — confirmed). DV: `dv/tests/test_amber_snoop_resp.py`, green at gate/func/full on both geometries; the FULL level is a 3000-snoop randomized soak over an evolving line-state model with per-transaction ACE compliance checks, not a token rerun.

13.1 Overview

`l1_snoop_resp` is the ACE-shaped snoop responder. It wraps the house `axi4ace_snoop_slave` module and adds the small state machine that sequences CR and CD for each snoop transaction. Inside `l1_core` the snoop is bus-agnostic; only `l1_snoop_resp` speaks ACE.

13.2 External Interface

The external side is the standard `axi4ace_snoop_slave` port:

Signal	Direction	Width	Purpose
`m_axi_acaddr`	input	`ADDR_WIDTH`	Snoop address
`m_axi_acsnoop`	input	4	Snoop type
`m_axi_acprot`	input	3	Snoop protection
`m_axi_acvalid`	input	1	AC valid
`m_axi_acready`	output	1	AC ready
`m_axi_crresp`	output	5	CR response
`m_axi_crvalid`	output	1	CR valid
`m_axi_crready`	input	1	CR ready
`m_axi_cddata`	output	`DATA_WIDTH`	CD data
`m_axi_cdlast`	output	1	CD last
`m_axi_cdvalid`	output	1	CD valid
`m_axi_cdready`	input	1	CD ready

13.3 Internal Interface

The core-facing side is a bus-agnostic handshake:

Signal	Direction	Meaning
`ctrl_snoop_req`	output to `l1_control`	Valid snoop address + type.
`ctrl_snoop_addr`	output	Snoop line address.
`ctrl_snoop_type`	output	Encoded snoop type.
`ctrl_snoop_ready`	input from `l1_control`	Control accepts snoop.
`ctrl_crresp`	input from `l1_control`	5-bit response.
`ctrl_cddata`	input from `l1_control`	CD beat data.
`ctrl_cdlast`	input from `l1_control`	CD last beat.
`ctrl_cdvalid`	input from `l1_control`	CD beat valid.
`ctrl_cdready`	output to `l1_control`	Adapter ready for CD beat.

13.4 Adapter State Machine

State	Meaning
`SR_IDLE`	Waiting for AC handshake.
`SR_LOOKUP`	Snoop accepted; waiting for `l1_control` CRRESP.
`SR_DATA`	Driving CD beats.
`SR_RESP`	Driving CR after CDLAST.

13.4.1 State transitions

1. `SR_IDLE` → `SR_LOOKUP`: AC handshake completes (`m_axi_acvalid && m_axi_acready`).
2. `SR_LOOKUP` → `SR_DATA`: `l1_control` returns `ctrl_crresp` and `CRRESP.DataTransfer` is 1.
3. `SR_LOOKUP` → `SR_RESP`: `l1_control` returns `ctrl_crresp` and `CRRESP.DataTransfer` is 0.

4. SR_DATA → SR_RESP: Last CD beat accepted (`m_axi_cdvalid && m_axi_cdready && m_axi_cdlast`).
5. SR_RESP → SR_IDLE: CR handshake completes (`m_axi_crvalid && m_axi_crready`).

13.5 CR-after-CDLAST Ordering

amber strengthens the protocol by completing all CD beats before asserting CR. This matches:

- IHI0022: the snoop response is required only after the final CD beat.
- The cocotb-framework 1.2.0 ACE compliance checker (`check_cr_order` / `check_cd_order`).
- The family convention recorded with onyx D4.

The CR channel is held low while the adapter is in SR_DATA. This makes the gather logic simple: one closed response at a time.

13.6 Snoop Type Decoding

`m_axi_acsnoop` carries the six IHI0022 snoop encodings used by amber:

Encoding	Type
ReadShared	Shared read snoop
ReadOnce	Single read snoop
ReadUnique	Exclusive read snoop
CleanShared	Clean shared snoop
CleanInvalid	Clean invalidate snoop
MakeInvalid	Invalidate snoop

CleanUnique and MakeUnique are transactions a master issues, not snoop encodings, so they do not appear on the AC channel. The CRRESP bits for each {state, snoop} pair are the authority of the HAS Table 3.0 and are captured as K-maps in the workbook.

Last Updated: 2026-10-06

14 amber_ace_issue: Coherent Transaction Mapping

Module: `amber_ace_issue.sv` **Location:** `projects/components/cache-ip/amber-mesi-l1/rtl/fub/` **Status:** RTL landed; this chapter is the micro-architecture contract the RTL implements.

14.1 Overview

`amber_ace_issue` maps cache-side events to the onyx D2 coherent-transaction subset. It lives only in the `amber_ace_top` rig; the pair-rig `amber` top does not instantiate it. The module sits between `l1_control` / `l1_fill` / `l1_drain` and the `axi4ace_master_rd` / `axi4ace_master_wr` wrappers.

14.2 Event-to-Transaction Mapping

Table 2.8.1: Cache events to ACE transactions

Cache event	ACE transaction	Snoop field	Notes
Read miss, shared intent	`ReadShared`	`fub_axi_arsnoop[3:0]`	Line will be installed in S.
Read miss, exclusive intent	`ReadUnique`	`fub_axi_arsnoop[3:0]`	Line will be installed in E.
Write to Shared line	`CleanUnique`	`fub_axi_awsnoop[2:0]`	Upgrades S → M without data transfer.
Write to Invalid line (whole line)	`MakeUnique`	`fub_axi_awsnoop[2:0]`	Invalidates other copies; line installed in M.
Dirty eviction	`WriteBack`	`fub_axi_awsnoop[2:0]`	Whole-line write-back with data.
Clean eviction	`Evict`	`fub_axi_awsnoop[2:0]`	No data; notifies interconnect.

14.3 Interface

14.3.1 From l1_control / fill / drain

Signal	Direction	Meaning
ace_rd_req	input	Read transaction request.
ace_rd_type	input	Encoded event type (shared read / unique read).
ace_rd_addr	input	Line-aligned address.
ace_rd_len	input	Burst length.
ace_wr_req	input	Write transaction request.
ace_wr_type	input	Encoded event type (CleanUnique / MakeUnique / WriteBack / Evict).
ace_wr_addr	input	Line-aligned address.
ace_wr_len	input	Burst length.

14.3.2 To axi4ace_master_rd / axi4ace_master_wr

amber_ace_issue drives the fub_axi_* ports of the ACE master wrappers. It supplies arsnoop/awsnoop and the standard AXI address/data fields. The wrappers auto-pulse RACK/WACK (deprecated per ruling 2026-10-11 (Sean); removal tracked as amber A5).

14.4 CleanUnique / MakeUnique Data Behavior

- CleanUnique is an AW-only transaction (no W data). It tells the interconnect to invalidate other copies so the local write can proceed. l1_control performs the actual data write into l1_data_array and promotes the line to M.
- MakeUnique is also an AW-only transaction for the full-line case. It invalidates other copies and grants exclusive ownership; the local line is installed or promoted to M.

- `WriteBack` is an AW+W burst with the dirty victim data.
- `Evict` is an AW-only transaction with no data.

14.5 Timing

`amber_ace_issue` is combinational for address/snoop mapping. It does not add pipeline stages; the timing path is from `l1_control` through this module into the `axi4ace_master_*` skid buffers.

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15 l1_cpu_frontend and l1_monlite

Modules: `l1_cpu_frontend.sv`, `l1_monlite.sv` **Location:** `projects/components/cache-ip/common-ip/rtl/fub/` **Status:** RTL landed; this chapter is the micro-architecture contract the RTL implements.

15.1 l1_cpu_frontend

15.1.1 Purpose

`l1_cpu_frontend` is the GAXI slave that accepts CPU requests and returns responses. Because the cache is blocking, it latches one request at a time and does not accept a new request until `l1_control` returns `req_ready`.

15.1.2 Request payload

```
CPU_REQ_W = ADDR_WIDTH + 1 + BUS_WIDTH/8 + BUS_WIDTH
```

Packed as `{addr[ADDR_WIDTH-1:0], we, be[BUS_WIDTH/8-1:0], wdata[BUS_WIDTH-1:0]}`.

15.1.3 Response payload

```
CPU_RSP_W = BUS_WIDTH
```

Packed as `{rdata[BUS_WIDTH-1:0]}`.

15.1.4 Handshake

Signal	Direction	Meaning
`cpu_req_wr_valid`	input	CPU has a request.
`cpu_req_wr_ready`	output	Frontend accepts request this cycle.
`cpu_req_wr_data`	input	Request payload.
`cpu_rsp_rd_valid`	output	Response ready.
`cpu_rsp_rd_ready`	input	CPU accepts response.
`cpu_rsp_rd_data`	output	Response payload.

`cpu_req_wr_ready` is asserted only when `l1_control` is in `CTRL_IDLE` and no snoop priority stall is active. Once a request is accepted, the front end latches it and holds it until `l1_control` returns the response.

15.1.5 Replay path

On a miss, `l1_control` re-drives the original latched request into its own hit logic in `CTRL_REPLAY`. The front end does not see a new GAXI transaction; the response is generated from the same latched request.

15.2 l1_monlite

15.2.1 Purpose

`l1_monlite` is the drop-and-count MonBus observer. It taps the internal signals of `l1_core` and emits 128-bit MonBus event packets. It never stalls the observed path; under sustained backpressure it drops packets and counts the loss.

15.2.2 Observation points

Event	Emit point	Payload highlights
hit	`CTRL_HIT_RD` / `CTRL_HIT_WR`	set, way, old/new state
miss	`CTRL_MISS_VICTIM`	set, way, miss class
snoop	snoop accepted	snoop type, hit/miss, response class
eviction	`CTRL_MISS_VICTIM` dirty victim	clean/dirty, set, way
MESI transition	state update	old state, new state, cause
fill start	`CTRL_MISS_FILL`	address, direction
fill end	`CTRL_FILL_WRITE`	address
drain start	`CTRL_MISS_DRAIN`	address
drain end	drain done	address

15.2.3 Packet format

Standard MonBus 128-bit packet with house UNIT/AGENT ids. The exact field layout follows the shared `monitor_common_pkg` spec; the MAS does not invent a new layout.

15.2.4 Drop-and-count

If `mon_ready` is low when a packet is ready, the packet is dropped and a saturating drop counter increments. The drop count is reported as an `EVENT_DROPPED` packet when the bus next has room, exactly like the STREAM monitor-lite behavior.

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16 CPU GAXI Request/Response Timing

16.1 Channel Structure

The CPU-side port is two house-GAXI streams:

Table 3.1.1: CPU GAXI channels

Signal	Direction	Width	Payload
`cpu_req_wr_valid`	in	1	request valid
`cpu_req_wr_ready`	out	1	request accepted
`cpu_req_wr_data`	in	`CPU_REQ_W`	packed {addr, we, be, wdata}
`cpu_rsp_rd_valid`	out	1	response valid
`cpu_rsp_rd_ready`	in	1	response accepted
`cpu_rsp_rd_data`	out	`CPU_RSP_W`	packed {rdata}

16.2 Request Timing

Because the cache is blocking, `cpu_req_wr_ready` is asserted only when `l1_control` is in `CTRL_IDLE` and no higher-priority snoop stall is active.

```
Cycle :  0   1   2   3   4   5
req_v :  1   0   0   0   0   0
req_r :  1   0   0   0   0   0   <- handshake at cycle 0
state :  L0  L1  HR  HR  RSP RSP
```

- Cycle 0: request accepted; `l1_cpu_frontend` latches it.
- Cycle 1: `CTRL_LOOKUP` drives tag-array port A.
- Cycle 2–3: hit service (`CTRL_HIT_RD` or `CTRL_HIT_WR`).
- Cycle 4–5: response valid until accepted.

On a miss, `cpu_req_wr_ready` stays low from cycle 1 through the entire miss sequence. The front end does not accept a new request until the replay completes.

16.3 Response Timing

`cpu_rsp_rd_valid` is asserted when `l1_control` enters the response state. The data is held stable until `cpu_rsp_rd_ready` is high. Because the response is single-beat and the front end blocks, backpressure is expected to be rare, but the handshake is still honored.

16.4 Write-Allocate Fills

A write miss triggers a fill, a merge of the CPU write data with the fill data, and then a replay of the write as a hit. The GAXI slave sees only the original single-beat write request and the eventual single-beat write response; the read-modify-write sequence is internal to `l1_control`.

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17 Fabric AXI4 Master Timing

The amber pair-rig top uses the house axi4_master_rd and axi4_master_wr wrappers for plain AXI4 memory access. This chapter documents the cycle-level handoff between l1_fill / l1_drain and those wrappers.

17.1 Read (Fill) Timing

l1_fill drives the fub_axi_* side of axi4_master_rd:

```
Cycle :  0   1   2   3   4   5   6
AR_v  :  1   0   0   0   0   0   0
AR_r  :  1   0   0   0   0   0   0   <- AR handshake
R_v   :  0   0   0   1   1   1   0
R_r   :  0   0   0   1   1   1   0   <- R beats accepted
last  :  0   0   0   0   0   1   0
```

- ARLEN = FILL_BEATS - 1, ARSIZE = log2(BUS_WIDTH/8), ARBURST = INCR.
- RREADY is asserted as long as l1_data_array port A can accept the beat.
- Each accepted R beat is written to l1_data_array and the pending-fill bypass register is updated.
- fill_done pulses one cycle after RLAST is accepted.

17.2 Write (Drain) Timing

l1_drain drives the fub_axi_* side of axi4_master_wr:

```
Cycle :  0   1   2   3   4   5   6
AW_v  :  1   0   0   0   0   0   0
AW_r  :  1   0   0   0   0   0   0   <- AW handshake
W_v   :  0   1   1   1   0   0   0
W_r   :  0   1   1   1   0   0   0   <- W beats accepted
last  :  0   0   0   1   0   0   0
B_v   :  0   0   0   0   0   1   0
B_r   :  0   0   0   0   0   1   0   <- B handshake
```

- AWLEN = FILL_BEATS - 1, AWSIZE = log2(BUS_WIDTH/8), AWBURST = INCR.
- WSTRB is all-1s for every beat.
- drain_done pulses one cycle after the B handshake.

17.3 Single Outstanding

There is at most one AR transaction and one AW transaction in flight at any time because the blocking pipeline does not launch a new miss until the previous one completes. No reorder buffer is needed inside `l1_fill` or `l1_drain`.

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18 Fabric ACE Master Timing

The amber_ace onyx-rig top uses the house axi4ace_master_rd and axi4ace_master_wr wrappers. The timing is identical to the AXI4 case except for the ARSNOOP/AWSNOOP fields, the constant AxDOMAIN/AxBAR fields, the 4-bit RRESP, and the auto-pulsed RACK/WACK outputs (deprecated per ruling 2026-10-11 (Sean); removal tracked as amber A5).

18.1 Domain, Barrier and Read-Response Fields

Every amber transaction is coherent and the onyx rig has one shareability domain, so amber_ace_top drives ARDOMAIN and AWDOMAIN to 0b10 (Outer Shareable), an encoding IHI 0022H Tables D3-7 and D3-8 permit for every Table 4.2 transaction type. Inner Shareable would also be legal on ACE, but ACE5 deprecates it. amber issues no barrier transactions (onyx D8), so ARBAR and AWBAR are 0b00, a normal access that respects barriers.

RRESP is 4 bits on an ACE read (IHI 0022H §D2.1.3, Table D3-14). RRESP[1:0] reaches the fill engine as before. RRESP[3:2] (IsShared, PassDirty) is honoured since issue #106 (commit 2594d73bb): the 4-bit response travels into l1_fill, and FILL_WRITE installs **M** when PassDirty = 1 (S otherwise) — a received dirty line keeps its write-back responsibility instead of being silently cleaned. The default install state still comes from the request class: ReadShared installs S and ReadUnique installs M. To make the PassDirty contract reachable, read misses now issue ReadNotSharedDirty (the IHI 0022H transaction defined for exactly this MESI case); the pre-#106 worry — a manager passing PassDirty = 1 on a plain ReadShared — is now handled rather than relied against. The port contract is stated here: l1_fill receives fill_pass_dirty from the read transport and forwards the installed state to l1_control.

18.2 Read Transactions

Cache event	ACE transaction	fub_axi_arsnoop
read miss, shared intent	ReadNotSharedDirty	ReadNotSharedDirty encoding (0x3) — issue #106
read miss, exclusive intent	ReadUnique	ReadUnique encoding (0x7)
upgrade: write to Shared line	CleanUnique	CleanUnique encoding (0xB) — AR channel, issue #106
upgrade: write to Invalid line (whole line)	MakeUnique	MakeUnique encoding (0xC) — AR channel, issue #106

RACK is auto-pulsed one cycle after RLAST is accepted (deprecated per ruling 2026-10-11 (Sean); removal tracked as amber A5). `amber_ace_issue` supplies `fub_axi_arsnoop`; `l1_fill` supplies address, length, size, and burst.

18.3 Write Transactions

Cache event	ACE transaction	`fub_axi_awsnoop`
dirty eviction	`WriteBack`	`WriteBack` encoding
clean eviction	`Evict`	`Evict` encoding

WACK is auto-pulsed one cycle after the B handshake (deprecated per ruling 2026-10-11 (Sean); removal tracked as amber A5). Since issue #106 (commit `2594d73bb`), `CleanUnique` and `MakeUnique` are **AR-channel** transactions with the spec encodings (0xB/0xC) — they share the read transport and carry no W data. `WriteBack` is an AW+W burst with the dirty victim data; `Evict` is AW-only with no W data.

18.4 Timing Example: ReadNotSharedDirty Fill

```
Cycle :  0   1   2   3   4   5   6
AR_v  :  1   0   0   0   0   0   0
AR_r  :  1   0   0   0   0   0   0   <- AR handshake,
ARSNOOP=ReadNotSharedDirty (0x3, issue #106)
R_v   :  0   0   0   1   1   1   0
R_r   :  0   0   0   1   1   1   0
last  :  0   0   0   0   0   1   0
rack  :  0   0   0   0   0   0   1   <- auto-pulsed
```

18.5 Timing Example: WriteBack Drain

```
Cycle :  0   1   2   3   4   5   6
AW_v  :  1   0   0   0   0   0   0
AW_r  :  1   0   0   0   0   0   0   <- AW handshake,
AWSNOOP=WriteBack
W_v   :  0   1   1   1   0   0   0
W_r   :  0   1   1   1   0   0   0
last  :  0   0   0   1   0   0   0
B_v   :  0   0   0   0   0   1   0
B_r   :  0   0   0   0   0   1   0
wack  :  0   0   0   0   0   0   1   <- auto-pulsed
```

Last Updated: 2026-10-11

19 Snoop AC/CD/CR Timing and Ordering

19.1 External ACE Snoop Port

Signal	Direction	Purpose
`m_axi_acvalid` / `m_axi_acready`	in / out	AC handshake
`m_axi_acaddr`	in	snoop line address
`m_axi_acsnoop`	in	snoop type
`m_axi_cdvalid` / `m_axi_cdready`	out / in	CD handshake
`m_axi_cddata`	out	snoop data beat
`m_axi_cdlast`	out	last CD beat
`m_axi_crvalid` / `m_axi_crready`	out / in	CR handshake
`m_axi_crresp`	out	5-bit response

19.2 Transaction Sequence

```
Cycle :  0   1   2   3   4   5   6   7
AC_v  :  1   0   0   0   0   0   0   0
AC_r  :  1   0   0   0   0   0   0   0   <- AC handshake
lookup:      L0  L1
CD_v  :  0   0   0   1   1   1   0   0
CD_r  :  0   0   0   1   1   1   0   0
last  :  0   0   0   0   0   1   0   0
CR_v  :  0   0   0   0   0   0   1   0
CR_r  :  0   0   0   0   0   0   1   0   <- CR after CDLAST
```

- Cycle 0: AC handshake. The snoop address and type are captured.
- Cycles 1–2: `l1_control` performs tag lookup on port B. If the line is M/E and the snoop requires data, it reads the data array port B.
- Cycles 3–5: CD beats are driven and accepted.
- Cycle 6: CR is asserted after `CDLAST` has been accepted.
- Cycle 7: CR handshake completes; the adapter returns to idle.

19.3 CR-After-CDLAST Rule

`l1_snoop_resp` holds `m_axi_crvalid` low until the last CD beat has been accepted. This is stronger than the minimum IHI0022 requirement but matches the family convention and the cocotb-framework 1.2.0 compliance checker. It also makes the gather logic simple: the manager sees one closed response at a time.

19.4 Multiple Outstanding Snoops

The `axi4ace_snoop_slave` wrapper buffers up to `SKID_DEPTH_AC` snoops. However, `l1_control` processes them one at a time because it is a single FSM. Therefore the effective outstanding snoop count is 1 from the core's perspective; the adapter FIFO absorbs bursts of AC arrivals.

19.5 AC Ready Backpressure

`m_axi_acready` is the skid-buffer `wr_ready` from `axi4ace_snoop_slave`. It can be low when the AC skid is full. The manager must respect this; amber never drops a snoop address because the adapter buffers it.

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20 MonBus Event Emission Points

20.1 Packet Format

`l1_monlite` emits standard 128-bit MonBus packets plus a 64-bit side-band timestamp. The packet layout is the house `monitor_common_pkg` spec; the timestamp is sampled at the same cycle as the packet and held stable while `mon_valid` is high.

20.2 Emission Timing

Events are emitted as one-cycle pulses. The event class and payload are captured combinatorially from the control FSM state and registered at the emission point to avoid glitches.

Event class	Emit point	Payload
hit	`CTRL_HIT_RD` / `CTRL_HIT_WR`	set, way, state before/after
miss	`CTRL_MISS_VICTIM`	set, way, miss class
snoop	AC handshake accepted	snoop type, hit/miss flag
eviction	`CTRL_MISS_DRAIN` start	clean/dirty, set, way
MESI transition	tag-array state write	old state, new state, cause code
fill_start	`CTRL_MISS_FILL`	line address
fill_end	`CTRL_FILL_WRITE`	line address
drain_start	`CTRL_MISS_DRAIN`	line address
drain_end	`drain_done`	line address

20.3 Drop-and-Count

If `mon_ready` is low when `mon_valid` would assert, the packet is dropped and a saturating drop counter increments. The counter is reported as an `EVENT_DROPPED` packet when MonBus next has room. This is identical to the STREAM monitor-lite behavior and never stalls the observed path.

20.4 Backpressure Integrity

`l1_monlite` is a pure observer. Its FIFO state never gates the CPU GAXI port, the AXI4/ACE masters, or the snoop responder. This is the measurement-integrity rationale that chose `*_monlite` over the heavyweight `_mon` wrappers.

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21 MonBus Event Encodings and Emission Points

21.1 Overview

Because amber has no register block, the MonBus observation map replaces the traditional register-map chapter. Every software-visible event is emitted as a MonBus packet by `l1_monlite`.

21.2 Event Encoding

Table 4.1.1: MonBus event encodings

Event code	Name	Emitting module	Trigger
`AMBER_EV_HIT`	hit	`l1_monlite`	`CTRL_HIT_RD` or `CTRL_HIT_WR`
`AMBER_EV_MISS`	miss	`l1_monlite`	`CTRL_MISS_VICTIM`
`AMBER_EV_SNOOP`	snoop	`l1_monlite`	AC handshake accepted
`AMBER_EV_EVICT`	eviction	`l1_monlite`	dirty victim staged to `l1_victim`
`AMBER_EV_TRANSITION`	MESI transition	`l1_monlite`	tag-array state write
`AMBER_EV_FILL_START`	fill start	`l1_monlite`	`CTRL_MISS_FILL`
`AMBER_EV_FILL_END`	fill end	`l1_monlite`	`CTRL_FILL_WRITE`
`AMBER_EV_DRAIN_START`	drain start	`l1_monlite`	`CTRL_MISS_DRAIN`
`AMBER_EV_DRAIN_END`	drain end	`l1_monlite`	`drain_done`
`AMBER_EV_DROPPED`	dropped packet	`l1_monlite`	packet dropped due to backpressure

The exact numeric codes are assigned in `l1_pkg` and match the house `monitor_common_pkg` allocation rules.

21.3 Payload Fields

All packets carry:

- `agent_id` — house `MON_AGENT_ID` for amber.
- `unit_id` — sub-unit within amber (control, snoop, etc.).
- `event_code` — one of the codes above.
- `timestamp` — 64-bit side-band sampled at emission.

Event-specific payload:

Event	Payload fields
hit	`set`, `way`, `state_before`, `state_after`, `we`
miss	`set`, `way`, `miss_class`, `we`
snoop	`snoop_type`, `hit`, `response_class`
eviction	`set`, `way`, `dirty`, `line_addr`
transition	`set`, `way`, `old_state`, `new_state`, `cause`
fill_start / fill_end	`line_addr`
drain_start / drain_end	`line_addr`
dropped	`dropped_count`

21.4 Miss-Class Encoding

Miss classes follow the `cache_sim` computation:

Code	Class	Meaning
`AMBER_MISS_COMPULSORY`	compulsory	First access to this line.
`AMBER_MISS_CAPACITY`	capacity	Line was evicted by capacity pressure.
`AMBER_MISS_CONFLICT`	conflict	Line was evicted by mapping conflict.

The golden model supplies the class; the RTL records what it is told. When the model is not present, the class is recorded as `AMBER_MISS_UNKNOWN`.

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22 Drop-and-Count Behavior and monbus_tally_axil

22.1 Drop-and-Count Policy

`l1_monlite` is drop-and-count, not stall-and-backpressure. This preserves measurement integrity: the observer cannot perturb the cache hit/miss timing it is trying to measure.

When a packet is emitted and `mon_ready` is low, the packet is discarded and a saturating counter increments. When `mon_ready` next goes high, if the counter is non-zero an `AMBER_EV_DROPPED` packet is emitted with the count, and the counter clears.

22.2 Counter Width

The drop counter is 8 bits. The probability of 256 consecutive drops in a correctly sized system is negligible; if it occurs, the counter saturates and the overflow is visible in the dropped packet.

22.3 monbus_tally_axil Integration

Amber's MonBus connects to the shared `monbus_tally_axil` agent exactly like STREAM and every other component in the repo. The agent:

1. Accepts `mon_valid`/`mon_ready`.
2. Parses the 128-bit packet using `monitor_common_pkg`.
3. Maintains per-event-class counters in AXI-Lite addressable registers.
4. Reports dropped-packet counts.

22.4 Software View

Software reads event counters through the `monbus_tally_axil` AXI-Lite slave. There is no APB CSR inside amber. The counter addresses and field layouts are defined by the tally agent, not by amber.

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23 What the K-maps Prove

23.1 Methodology

The kmap workbook (`gen_amber_contracts_kmaps.py`) applies the STREAM-derived methodology:

1. Identify every important combinational decision signal.
2. List axes as `(name, expression, cite)` triples.
3. State `depends_only_on` evidence that justifies the axis list.
4. Add checked `relations` that mark unreachable input combos as don't-cares.
5. Derive the minimal SOP mechanically with Quine-McCluskey.
6. Diff the derived cover against the RTL SOP when RTL exists.

Because amber has no RTL yet, the workbook is a **pre-RTL contract**. The derived SOP is the logic the future RTL must implement. `rtl_sop` is intentionally absent in the workbook; when RTL lands the diff step produces one of three verdicts.

23.2 The Three Verdicts

The verdict logic is implemented in `bin/kmaps/minimize.py`:

Verdict	Meaning	Action
IDENTICAL	The RTL SOP matches the derived minimal cover.	No change needed; the RTL is already minimal.
RTL-REDUNDANT	The RTL SOP includes extra literals or terms but produces the same truth table.	Document why the redundancy exists (timing, readability, defensive coding).
RTL-DIFFERS	The RTL SOP produces a different truth table than the derived cover.	Investigate: bug, unstated invariant, or missing relation.

23.3 Task 14 K-map Verdict Sweep (2026-10-11)

All RTL for the amber MESI L1 control plane has landed. The workbook contains three K-map sheets; there is no MonBus K-map sheet (MonBus appears only as the `MonBus out` contract sheet).

Verdicts below are evaluated against the landed RTL in `projects/components/cache-ip/amber-mesi-l1/rtl/`.

23.3.1 Verdict summary

Sheet	Rows	IDENTICAL	RTL-REDUNDANT	RTL-DIFFERS
K-maps snoop CRRESP	DT, PD, IS, WU	0	4	0
K-maps snoop next-state	next_state[1], next_state[0]	0	2	0
K-maps amber control	start_drain, start_fill, replay_now	3	0	0
Total	**9**	**3**	**6**	**0**

No NEW differences were found that are not already covered by the known errata pinned in the plan and recorded in `02_rtl_diff.md`.

23.3.2 K-maps snoop CRRESP (l1_snoop_kmap / l1_pkg)

Authority: HAS Table 3.0, implemented in `cache-ip/common-ip/rtl/includes/l1_pkg.sv` function `amber_snoop_crresp` and wrapped by `cache-ip/common-ip/rtl/fub/l1_snoop_kmap.sv`.

Row	Verdict	Rationale / Review Focus
DT (DataTransfer)	**RTL-REDUNDANT**	Truth table matches the HAS Table 3.0 1-cells exactly for all reachable {state,snoop} combinations. The RTL expresses the map as a readable `case` decode over `{state,snoop}` rather than the QM-derived minimal SOP; reserved snoop codes (6/7) and the reserved `STATE_0` code are folded to the safe default `0` instead of being exploited as don't-cares. This is defensive coding: the adapter (`l1_snoop_resp`) routes illegal 4-bit ACSNOOP encodings to the reserved 3-bit code

Row	Verdict	Rationale / Review Focus
		`3'b110`, and the kmap safe default prevents spurious data transfer on those paths (Review Focus 10: recorded gaps stay recorded).
PD (PassDirty)	**RTL-REDUNDANT**	Same case-decode implementation and truth-table match as DT. The RTL produces PD=1 only for M-state snoops that transfer data, matching Table 3.0.
IS (IsShared)	**RTL-REDUNDANT**	Same case-decode implementation and truth-table match as DT. Notably, M + CleanInvalid returns IS=1 per the corrected Table 3.0 (this was the discrepancy caught and fixed during the first-verdict TB bring-up).
WU (WasUnique)	**RTL-REDUNDANT**	Same case-decode implementation and truth-table match as DT. WU=1 only for E-state data-transfer snoops.

23.3.3 K-maps `snoop next-state` (l1_snoop_kmap / l1_pkg)

Authority: HAS Table 3.0 next-state column, implemented in `cache-ip/common-ip/rtl/includes/l1_pkg.sv` function `amber_snoop_next_state` and wrapped by `cache-ip/common-ip/rtl/fub/l1_snoop_kmap.sv`.

Row	Verdict	Rationale / Review Focus
next_state[1]	**RTL-REDUNDANT**	Truth table matches Table 3.0 for all reachable {state,snoop} combinations. The RTL uses the same case decode as `amber_snoop_crresp`; reserved encodings fold to `STATE_I`.
next_state[0]	**RTL-REDUNDANT**	Same as next_state[1].

23.3.4 K-maps `amber control` miss-path decision (l1_control)

Authority: MAS ch02_blocks/01, implemented in `cache-ip/common-ip/rtl/fub/l1_control.sv` (victim-drain / fill-start arbitration).

Row	Verdict	Rationale / Review Focus
start_drain	**IDENTICAL**	RTL implements `kmap_start_drain = kmap_victim_dirty && !bypass_match`, which is exactly the workbook's proposed cover evaluated

Row	Verdict	Rationale / Review Focus
		in CTRL_MISS_VICTIM where hit == 0 by construction. The dirty-victim path gathers the line over FILL_BEATS port-A cycles and stages it into l1_victim (Review Focus 2: victim-buffer overflow is avoided by the single-outstanding invariant plus the leaf's !victim_valid_q guard; Review Focus 3: PassDirty/WB ordering is preserved because the victim buffer owns the line until drain_done).
start_fill	**IDENTICAL**	RTL implements kmap_start_fill = ! kmap_victim_dirty && !bypass_match, exactly the proposed cover.
replay_ now	**IDENTICAL**	RTL implements kmap_replay_now = bypass_match, exactly the proposed cover. On the CPU miss path bypass_match is constant-0 by construction (the pending-fill bypass lifetime is exactly CTRL_MISS_FILL..CTRL_FILL_WRITE of the single outstanding transaction), so this term is defensive; it is reachable on the snoop decision path when a snoop matches a pending fill (Review Focus 1: snoop-during-fill race resolved by answering from the bypass at the post-fill pf_state, with CD beats gated by pf_data_valid).

23.3.5 MonBus / address-decode sheets

The workbook does not contain K-map sheets for MonBus events or address decode. MonBus out is a contract sheet, not a K-map. The MonBus event map is verified by dv/tests/test_amber_monlite.py against l1_pkg event codes and the MAS ch04 Table 4.1.1 emit points; Review Focus 5 (monlite perturbation) is embodied by cache-ip/common-ip/rtl/fub/l1_monlite.sv's drop-and-count path, which never stalls the observed signals.

23.4 Historical Verdicts

23.4.1 Control-Sheet Verdict (2026-10-07, updated Task 4)

`l1_control` landed (Task 3, CPU path; Task 4, snoop service). Verdict for the `K-maps amber control` miss-path sheet (axes `{hit, victim_dirty, pending_bypass_match}`, outputs `{start_drain, start_fill, replay_now}`):

- **Truth-table equivalence: PASS at gate/func/full, both geometries** (tiny s16w2 + default s128w4). The RTL implements the workbook's PROPOSED cover literally, evaluated in `CTRL_MISS_VICTIM` where `hit == 0` by construction: `start_drain = victim_dirty & !pending_bypass_match`, `start_fill = !victim_dirty & !pending_bypass_match`, `replay_now = pending_bypass_match`. The two reachable cells are exercised by directed dirty/clean-victim misses and by the 10k-transaction randomized oracle-lockstep suite (`dv/tests/test_amber_control.py`).
- **`pending_bypass_match` axis (2026-10-07 Task 4 update): now stimulus-reachable — driven on the snoop decision path.** With `CTRL_SNOOP` service live, the same pf registers feed the snoop reference-state resolution (bypass match → post-fill `pf_state` + `pf_data_valid`-gated CD beats). Directed classes `SnoopPendingFillBypass` (pre-RLAST snoop, CD stall until a late beat), `SnoopOtherLineMidFill` / `SnoopPostCommitApplies` (pf no-match → port-B installed-state decode), `ImStepPendingClearCorner` (repeat snoop with the invalidation pending), and `UpgradeNoBypassArm` (upgrade in flight: pf unarmed, snoop answered from the installed S entry) exercise both polarities of the axis plus the every-cycle `BypassNeverAnswersInvalid` invariant. On the miss path itself the axis remains constant-0 **by construction** — the pf register's lifetime is exactly `CTRL_MISS_FILL..CTRL_FILL_WRITE` of the single outstanding transaction and fills re-arm only after `CTRL_FILL_WRITE` clears it, so `CTRL_MISS_VICTIM` entry is exclusive with an armed bypass; the killed-upgrade re-fetch (`ImStepPendingClearCorner`, `UpgradeKilledByInvalidatingSnoop`) drives the full LOOKUP→MISS_VICTIM→MISS_FILL→FILL_WRITE→REPLAY re-lookup sequence with the axis at its constant-0 value. `replay_now` stays defensive-only.
- **SOP-literal diff against the QM-derived covers: NOT CHECKED**, same rationale as the snoop sheets (the RTL expresses the cover as named

decision wires, not minimized literals; the meaningful diff is the truth-table one).

23.4.2 First-Verdict Status (2026-10-06)

The first RTL slice landed: `l1_pkg` (the Table 3.0 decode functions `amber_snoop_crresp` / `amber_snoop_next_state`) and its module wrapper `l1_snoop_kmap`. Verdicts for the six snoop sheets (the four CRRESP maps and the two next-state maps):

- **Truth-table equivalence: PASS at gate/func/full.** The TB (`dv/tests/test_amber_snoop_kmap.py`) carries an independent copy of HAS Table 3.0 and exhausts the reachable space at gate, adds the reserved encodings at func, and the whole 3-bit x 3-bit space at full. One real discrepancy was caught and resolved in the TB's favor of the documentation: Modified + CleanInvalid carries `IsShared=1` per Table 3.0.
- **SOP-literal diff against the QM-derived covers: NOT CHECKED, deferred.** The RTL implements the table as a case decode, not minimal SOP literals; the meaningful diff is the truth-table one above. Supplying `rtl_sop=` and re-deriving is deferred until `l1_snoop_resp` integrates this decode — if the integration restructures the logic, the literal diff would be repeated work.

Last Updated: 2026-10-11 (Task 14 verdict sweep)

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24 RTL Landing Diff Procedure

24.1 When RTL Lands

After the first RTL commit for a module, update the corresponding sheet in `gen_amber_contracts_kmaps.py`:

1. Add citations to the RTL file and line numbers in the `CITES` registry.
2. Replace `rtl_sop=None` with the actual RTL expression as a sum-of-products string.
3. Re-run the generator.
4. Review each sheet's verdict.

First landing (2026-10-06): the snoop CRRESP / next-state decode exists as `l1_pkg` functions + `l1_snoop_kmap`. See 01_kmap_verdicts.md for why its sheets record truth-table PASS but the SOP-literal diff is classified RTL-REDUNDANT; the procedure above applies to every sheet still at `rtl_sop=None`.

24.2 Diff Checklist

For every sheet that reports **RTL-DIFFERS**:

- Is the difference a deliberate redundancy? Re-label as RTL-REDUNDANT and add a note.
- Is an unstated invariant doing work? Add it as a checked `relation` and re-derive.
- Is the RTL expression wrong? File a bug and fix it.

For every sheet that reports **RTL-REDUNDANT**:

- Does the redundancy serve timing closure? Keep it and document.
- Does the redundancy serve readability? Keep it if the cost is small.
- Is it accidental? Simplify the RTL.

24.3 Regression Rule

The generator is part of the CI check for amber. Any RTL edit that moves or changes a cited expression must cause the generator to fail loudly, not silently publish a stale map. This is enforced by the `CITES` registry in `gen_amber_contracts_kmaps.py`.

24.4 Task 14 RTL-vs-MAS Diff Notes (2026-10-11)

The following differences between the landed RTL and the MAS text were evaluated during the Task 14 kmap verdict sweep. All of them map onto the known errata pinned in the plan; **no NEW differences were found** in the kmap-relevant scope.

#	MAS text / figure	Landed RTL	Verdict class	Known erratum
1	ch01 and ch03 describe tag/data arrays as “built from `sdpram_core`”	Arrays are per-way inferred RAMs in the `sdpram_core` internal idiom; the literal `sdpram_core` module is a FUB/AXI burst slave and is the wrong shape for a same-cycle multi-way tag compare. The Task 9 ruling keeps the landed arrays and satisfies D11’s queue requirement with the two `gaxi_fifo_sync` instances.	RTL-DIFFERS (documentation wording)	MAS ch01↔ch03 `sdpram_core` wording contradiction
2	ch02 port list / handshake table uses	`l1_control` and	RTL-DIFFERS (naming)	ch02 `ctrl_snoop_*` →

#	MAS text / figure	Landed RTL	Verdict class	Known erratum
	ctrl_snoop_crresp, ctrl_snoop_cddata, ctrl_snoop_cdlast	l1_snoop_resp use ctrl_crresp, ctrl_cddata, ctrl_cdlast, plus the streaming handshake ctrl_cdvalid/ctrl_cdready.		landed ctrl_* names (RTL wins)
3	ch02/05 shows a single-cycle victim_load that loads a full line	The data-array port is BUS_WIDTH wide (MAS ch03), so a full line cannot be loaded in one cycle. l1_control gathers the dirty victim line over FILL_BEATS port-A read cycles and uses ctrl_victim_load as the final-beat strobe.	RTL-DIFFERS (timing)	ch02 single-cycle victim_load → gather over FILL_BEATS with final-beat strobe (D-5)
4	ch02 FSM table lists CTRL_IDLE, CTRL_LOOKUP, CTRL_HIT_RD,	l1_pkg and l1_control implement an additional	RTL-DIFFERS (FSM states)	ch02 FSM table missing CTRL_INIT (D-3)

#	MAS text / figure	Landed RTL	Verdict class	Known erratum
	CTRL_HIT_WR, CTRL_MISS_VICTIM, CTRL_MISS_DRAIN, CTRL_MISS_FILL, CTRL_FILL_WRITE, CTRL_REPLAY, CTRL_SNOOP, CTRL_ERROR	stable state CTRL_INIT for the post-reset init walk that writes STATE_I to every way before the first lookup.		
5	ch01/02 top-level port list omits the coherence sideband	amber_top and amber_ace_top expose the pair-rig coherence sideband coh_req_{valid,addr,type} (DECISION D-8).	RTL-DIFFERS (port list)	ch01/02 port list missing the coherence sideband
6	Plan / early testplans reference dv/tests/fub/	All amber unit and macro suites live in the flat dv/tests/ directory, consistent with the repo convention and the plan's DECISION D-13.	RTL-DIFFERS (paths)	dv/tests/fub/ → flat dv/tests/ paths

24.4.1 Design evolutions that are not kmap diffs

The following landed design choices are recorded here for completeness but do not change the truth table of any K-map sheet:

- **DECISION D-2 (leaf modules):** `l1_pending_fill_bypass` and `l1_victim` are built as leaf modules (not inline control logic) so Task 13 can prove them standalone. The port-level contract matches the MAS fields; the leaf boundary is an RTL-REDUNDANT structural choice.
- **DECISION D-6 (fill/drain queues):** `l1_fill` uses a `gaxi_fifo_sync` (DEPTH=4) to stage R beats so snoop-imposed port-A stalls never drop `fub_axi_rready`; `l1_drain` forwards victim beats directly because the `axi4_master_wr` wrapper's W skid absorbs backpressure. This changes the pre-RTL fill/drain handshake sketch but does not affect the control-path K-map.
- **DECISION D-9 (RANDOM policy):** `l1_repl` RANDOM uses an LFSR; `cache_sim` uses mulberry32. Exact bit-for-bit parity is impossible; the parity suite records this caveat. This is outside the K-map sheets.

Last Updated: 2026-10-11 (Task 14 RTL-vs-MAS diff sweep)

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25 gem5 FSM Cross-Check

25.1 Source

The Python reference model and the formal FSM oracles derive from the gem5 Ruby `MESI_Two_Level` SLICC tables, specifically `L1cache.sm`. The enumerated transient states in that file are the source for the `l1_pkg` state encoding and for the control FSM states in `l1_control`.

25.2 Cross-Check Mechanism

1. Extract every stable and transient state from `L1cache.sm`.
2. Map gem5 events to amber signals (CPU request, snoop type, fill response, drain done).
3. Replay a randomized trace through both the gem5-derived Python model and a future cycle-accurate RTL model.
4. Compare state transitions, CRRESP outputs, and next-state writes.

25.3 Scope

The cross-check covers:

- All M/E/S/I stable states.
- Transient states between stable states and pending responses.
- The six IHI0022 snoop types.
- The onyx D2 coherent issue subset (ReadShared, ReadUnique, CleanUnique, MakeUnique, WriteBack, Evict).

MOESI O-state is out of scope for v1.0; the 3-bit encoding reserves a code for it.

Last Updated: 2026-10-06

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26 Elaboration Configurations

26.1 Default Config

The default elaboration geometry is the PRD D1 center point:

Parameter	Value
SETS	128
WAYS	4
LINE_BYTES	64
BUS_WIDTH	64
REPL_POLICY	"LRU"
WRITE_POLICY	"wb_wa"
VICTIM_DEPTH	1

This is the configuration used for FPGA characterization, cache_sim parity, and the FULL cocotb grid.

26.2 Tiny Formal Config

The SymbiYosys control-layer proofs use a tiny geometry for tractability:

Parameter	Value
SETS	16
WAYS	2
LINE_BYTES	64
BUS_WIDTH	64
REPL_POLICY	"LRU"
WRITE_POLICY	"wb_wa"
VICTIM_DEPTH	1

The tiny config reduces tag-array depth and way count so that the formal tool can fully explore the control FSM and pending-fill bypass interactions.

26.3 Bring-Up Config

A minimal bring-up config may use:

Parameter	Value
SETS	4
WAYS	2
LINE_BYTES	32
BUS_WIDTH	32
REPL_POLICY	"FIFO"
WRITE_POLICY	"wt_na"
VICTIM_DEPTH	1

This is for initial module-level smoke tests only, not for characterization or formal proofs.

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27 DV Matrix Dimensions

27.1 Pattern-B Cocotb Grids

The primary DV layer is Pattern-B GATE/FUNC/FULL cocotb grids over the geometry × policy × rig matrix.

27.1.1 Dimensions

Dimension	Values
Geometry	default (128/4/64/64), tiny (16/2/64/64), bring-up (4/2/32/32)
Policy	LRU, FIFO, RANDOM, tree-PLRU
Rig	`amber` (pair-rig AXI4), `amber_ace` (onyx-rig ACE)
Write policy	wb_wa, wt_na (bring-up only)

27.1.2 Grid Coverage

Grid	Geometry	Policy	Rig	Purpose
GATE	tiny	LRU	amber	fast gate-level sanity
FUNC	default	LRU, FIFO, RANDOM	amber	functional regression
FUNC	default	LRU	amber_ace	coherent-transaction sanity
FULL	default	all four	both rigs	nightly regression + cache_sim parity
FULL	tiny	LRU	both rigs	formal companion runs

27.2 Formal Targets

SymbiYosys is scoped to the control layer at the tiny geometry:

- No stale data served after an external write (pending-fill bypass correctness).

- No deadlock or livelock under fair arbitration.
- Victim-buffer handoff correctness.
- `l1_snoop_resp` AC/CR/CD ordering.

27.3 Golden-Model Parity

For LRU, FIFO, and RANDOM, amber hit/miss/miss-class counts must match `cache_sim` at identical geometry and policy. tree-PLRU has no `cache_sim` golden model today; it is cross-checked by RTL self-checks and recorded as a future sim extension.

27.4 MonBus Coverage

Every event class in Chapter 4 must be visible on MonBus in simulation. The `_monlite` wrapper is present-vs-absent in the regression so the gate-cost delta is part of the evidence.

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28 amber — Product Requirements (DRAFT)

Version: 0.7 (2026-10-15) **Status:** all eleven decision rows record a name and a date. A decision becomes DECIDED when it records a name and a date, the way the reed-solomon PRD does it. v0.2: D8 decided — `*_monlite` observation wrappers (never `_mon` on the measured paths). v0.3: D4 decided — snoop transport is ACE-shaped (AC/CD/CR) per the onyx D7 port contract, family decision 2026-10-05. The snoop-filter (directory-lite) half of D4 stays deferred and is reopened as onyx D3. v0.4: D2 decided (Sean, 2026-10-06) — CPU-side interface is a **GAXI slave**. House BFM/monitor coverage and skid/FIFO plumbing already exist, and STREAM can attach as first real consumer without an adapter. v0.5 (Sean, 2026-10-06): D1, D5, D6, D7, D9, D10 decided in one working session — geometry 32 KiB/64 B/4-way center + a tiny 16-set/2-way formal config; write-back + write-allocate; plain MESI on a 3-bit state field; replacement {LRU, tree-PLRU, FIFO, RANDOM} with LRU default (tree-PLRU the timing fallback, no sim golden model — future sim extension recorded); verification = Pattern-B grids + control-layer SymbiYosys at tiny geometry + cache_sim parity on LRU/FIFO/RANDOM + FSM oracles derived from the gem5 Ruby MESI_Two_Level SLICC tables; first consumer = TB masters bring-up, pair rig the gated deliverable, STREAM attach the documented future hook. v0.6 (Sean, 2026-10-07): D3 and D11 decided, unblocking the cache-core RTL. D3 — memory-side is **AXI4 read/write masters on the house `axi4_master_rd`/`axi4_master_wr` wrappers (the stream/rapids pattern)**, observed through `*_monlite` per D8: the monlite decision needs monitored memory-side channels, the masters match the house BFM/monitor coverage that D2 chose GAXI for on the CPU side, and the pair rig's shared memory hangs off them through house fabric. The simple-SRAM-port bring-up shortcut was set aside as a rewrite-in-waiting. D11 — **DECIDED in the documented shared-primitives direction: `sdpram_core`-based tag/data stores (house no-bespoke-SRAM rule) + `gaxi_fifo_sync` house FIFOs for the pending/fill queues**; FPGA attributes per GLOBAL_REQUIREMENTS.md. v0.7 (2026-10-15): post-RTL reconciliation, no new D-rows. Shared core renamed `amber_*` → `l1_*` (commit 79dc18ccb) — D9's formal scope is now `l1_control` / `l1_snoop_resp` / `l1_pkg`. ACE issue #106 conformance (`2594d73bb`): read misses issue `ReadNotSharedDirty`, `PassDirty` is honoured (installs M), `CleanUnique`/`MakeUnique` move to the AR channel with the spec encodings — Table 2.8.1's event map is implemented that way. `PF_GUARD` pending-fill snoop guard added (`ab308d108`, amber BUG-002); `amber_top` sets it, standalone benches default it off. Genesys 2 pair-rig bring-up board-verified; bug board BUG-001/002/003 closed.

28.1 1. Purpose

A parameterised, blocking, snoopy MESI L1 data cache for the Sherpa research/teaching flow: correct against a software golden model, provable with SymbiYosys, observable through the MonBus monitor fabric, and characterised for FPGA cost like every other component in this repo. It is the first cache-ip IP and the deliberate baseline for jet, the lockup-free follow-on.

28.2 2. Background

A blocking cache freezes on a miss: the pending request (and everything behind it) waits for the fill to complete. That is poor for performance and excellent for correctness work — one outstanding transaction means no MSHR bookkeeping, no hit-under-miss hazards, no merge logic. MESI (Modified / Exclusive / Shared / Invalid) is the coherence layer on top: each line carries a state that says who may write it and who must be told when they hold a stale copy. Snoopy coherence broadcasts probes on a shared bus; every cache watches and answers.

Blocking and MESI are orthogonal, which is the design bet: amber nails the coherence protocol with trivial miss handling; jet keeps the protocol and upgrades the miss handling. The pair then answers the classic question with measured, on-board data: what does lockup-freedom actually buy?

Repo anchors the design must respect: resets on the `ALWAYS_FF_RST` macros, storage on the shared `sdpram_core` / gaxi FIFO primitives (no bespoke SRAM, no reset on memories), `[DEPTH]` array syntax, valid/ready streaming contracts, and the monitor/MonBus observation fabric from `rtl/amba/`.

28.3 3. Decisions that pick the design

#	Decision	Candidates	What it drives
D1	Cache geometry	**DECIDED 2026-10-06 (Sean): 32 KiB / 64 B lines / 4 ways (128 sets) as the default center, fully elaboration-parameterised (4–32 KiB, 32–64 B, 2–8 ways), plus a reserved tiny config — 16 sets / 2 ways / 64 B — for SymbiYosys tractability.** Board BRAM does not bind (~9 of 135 36Kb RAMs on Nexys A7 at	tag/data array dimensions, the whole DV matrix

#	Decision	Candidates	What it drives
		the default), so geometry serves the DV matrix and the proofs, not the board	
D2	CPU-side interface	**DECIDED 2026-10-06 (Sean): GAXI slave.** Chosen over the plain valid/ready native port and the AXI4 slave: house BFM/monitor coverage and skid/FIFO plumbing already exist, and STREAM can attach as first real consumer without an adapter. Must express single-beat reads, write-allocate fills, and (if write-through is chosen) store-without-allocate	front-end FSM, adapter surface, first consumer
D3	Memory-side interface	**DECIDED 2026-10-07 (Sean): AXI4 read/write masters on the house `axi4_master_rd`/`axi4_master_wr`**	fill/drain engines, burst behavior on misses

#	Decision	Candidates	What it drives
		wrappers (stream/rapids pattern), `*_monlite` observation per D8. Chosen over GAXI (D2 already spent that on the CPU side) and over the simple SRAM bring-up port (a rewrite-in-waiting); matches D8's monitored-paths requirement and the pair-rig shared-memory hook	
D4	Snoop transport	**DECIDED 2026-10-05 (Sean): ACE-shaped transport — amber implements the AC/CD/CR snoop channels exactly as onyx D7 defines them (family decision: cache-ip is ACE-shaped), staying bus-agnostic internally behind a thin adapter.** Chosen	the coherence bus, formal surface, gate count

#	Decision	Candidates	What it drives
		over a custom dedicated snoop bus and over snooping on the AXI4 fabric directly; the ACE-lite-shaped candidate was set aside because a snoop responder must drive CR responses and CD data — ACE-Lite carries no snoop channels at all. The snoop-filter (directory-lite) sub-question stays deferred, reopened as onyx D3.	
D5	Write policy	**DECIDED 2026-10-06 (Sean): write-back + write-allocate.** Write-through/no-allocate deletes the coherence traffic the research exists to measure — dirty snoop transfers (PassDirty), WriteBack drains, onyx's memory-read elision —	M-state logic, memory traffic, formal targets

#	Decision	Candidates	What it drives
		and leaves MESI's M state with no job; it stays available as a bring-up mode only	
D6	MESI variant	**DECIDED 2026-10-06 (Sean): plain MESI on a 3-bit state field.** MOESI's O-state is a later elaboration upgrade (encoding headroom reserved now, ~512 extra tag bits); its win — dirty→shared without a memory read — is a measurement for the pair rig, not a v1 requirement	state machine count, snoop response matrix
D7	Replacement policy	**DECIDED 2026-10-06 (Sean): policy set {LRU, tree-PLRU, FIFO, RANDOM}, default LRU.** LRU/FIFO/RANDOM match the cache simulator exactly (`POLICIES =`	policy engine, cross-check fidelity

#	Decision	Candidates	What it drives
		`['LRU','FIFO','RANDOM']`), so golden-model parity covers every policy both sides implement; **tree-PLRU is carried as the timing-tight fallback — the one policy with no sim golden model**, cross-checked by RTL self-checks today and a future cache_sim extension (recorded, not silently assumed). Policy stays an elaboration parameter so the sim↔RTL cross-check holds per policy	
D8	Observation	**DECIDED 2026-10-04 (Sean): `*_monlite` wrappers, not the heavyweight `_mon`.** MonBus taps for hits, misses (class split if the model supplies it),	perf counters, board capture, DV scoreboards

#	Decision	Candidates	What it drives
		snoops, evictions, and state transitions, observed through the `axi4_monlite` / `axil4_monlite` family — same 128-bit MonBus packets, UNIT/AGENT ids, and host tooling as `_mon`, ~1/5 the monitor gates, and a drop-and-count policy that never stalls the port. Rationale for a research cache: the observer must not perturb what it measures — a `_mon` wrapper that gates the port when its tracking tables fill would corrupt the very miss-latency numbers amber and jet exist to produce. The events D8 lists are emitted as MonBus packets and tallied by the standard agents	

#	Decision	Candidates	What it drives
		(`monbus_tally_axil`); heavyweight `_mon` stays available as a DV cross-check, never on the measured paths.	
D9	Verification strategy	**DECIDED 2026-10-06 (Sean): Pattern-B cocotb GATE/FUNC/FULL grids over the geometry × policy × rig matrix (ACE BFMs: cocotb-framework 1.2.0); SymbiYosys scoped to the control layer — `l1_control`, `l1_snoop_resp`, the pending-fill bypass, the victim-buffer handoff — proven at the tiny 16-set/2-way geometry; cache_sim trace-replay parity across LRU/FIFO/RANDOM. The Python reference model**	DV structure, formal areas, CI time

#	Decision	Candidates	What it drives
		and the FSM oracles derive from the gem5 Ruby `MESI_Two_Level` SLICC tables in `../References/gem5-ruby-protocols/` — `L1cache.sm`'s enumerated transient states are the derivation source for the `l1_pkg` state encoding, and every stable and transient transition is cross-checked against an executable protocol spec, not prose	
D10	First consumer	**DECIDED 2026-10-06 (Sean): standalone TB masters for bring-up; the gated deliverable is the pair rig — two ambers + shared memory, the coherence research rig.**	integration scope, what "done" means

#	Decision	Candidates	What it drives
		STREAM attach is documented as the future integration hook (D2's GAXI choice is what makes it cheap later); the onyx rig is sequenced per onyx D10 — onyx waits behind the pair rig	
D11	Data/tag arrays	**DECIDED 2026-10-07 (Sean), in the documented shared-primitives direction: `sdpram_core`-based tag/data stores** + **`gaxi_fifo_sync` house FIFOs for the pending/fill queues**; FPGA attributes per `../../../../GLOBAL_REQUIREMENTS.md`. House no-bespoke-SRAM rule applies	area/timing headroom, the no-bespoke-SRAM rule

28.4 4. Success criteria

1. **Golden-model parity.** Given the same trace + policy + geometry, amber's hit/miss/miss-class counts match cache_sim exactly across a regression trace suite.

2. **Proofs.** SymbiYosys: no stale data is ever served after an external write; the coherence protocol cannot deadlock under fair arbitration.
3. **Observation.** Every event class above visible on MonBus in simulation, and captured on at least one FPGA board.
4. **Cost.** Synthesis/implementation numbers (LUT/FF/RAM, fmax) on Nexys A7 and/or Genesys 2, recorded in `docs/` the way pumice characterised.

28.5 5. Non-goals

Non-blocking / lockup-free miss handling (jet), directory-based coherence, L2/L3 or multi-level inclusion policies, ECC on cache arrays, virtual memory and TLB interaction, multi-port CPU issue. Each is a future IP or a future decision, not an amber requirement.

28.6 6. Where the block sits

Research cache on the AMBA fabric: CPU-side slave (D2), memory-side AXI4 masters (D3), snoop port toward onyx / peer caches (D4 — ACE-shaped AC/CD/CR, decided 2026-10-05), MonBus observation (D8). Standalone-testable with TB masters before any consumer exists (D10).

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29 amber — Pre-Hardware Architecture Specification (Pre-HAS)

Version: 0.3 (draft, 2026-10-06) — **superseded by the full HAS (v1.0, same day)** once the PRD closed its rows (v0.5): the chapter book ratifies this sketch, and this file is kept as the decision-path record. **Status:** architecture sketch written while the PRD decisions are still OPEN. This document records **no decisions**. "PROPOSED" marks a working default put forward so the decision rows can be argued with concretely; a decision becomes DECIDED only when PRD.md records a name and a date. When the decisions close, each PROPOSED item here is either ratified into the full `amber_has` chapter book (the stream/andesite convention) or overridden and this file edited. v0.2 (2026-10-06): D2 landed in the PRD — CPU-side is a **GAXI slave**. F9 now answers probe-during-fill with a **pending-fill bypass** instead of an unbounded stall. Q1 answered — `amber_fabric` carries **both** rig attachments and the `RIG` elaboration parameter generate-selects. v0.3 (2026-10-06): Q1 re-answered by the owner — both rigs as **two tops**, `amber` (pair rig) and `amber_ace` (onyx rig), on a shared `amber_core`. No `RIG` parameter: the rig is whichever top you instantiate.

Governing documents: PRD.md (binding) · onyx PRD D7 (snoop-port contract) · jet PRD (what must stay constant) · GLOBAL_REQUIREMENTS.md · cache simulator (golden model)

29.1 1. Intent of this document

The PRD carries eleven decision rows and has closed three: **D2** (CPU-side is a GAXI slave), **D4** (snoop transport is ACE-shaped AC/CD/CR), and **D8** (observation through `*_monlite` wrappers only). The Pre-HAS makes the open rows concrete: for each, it sketches the feature the way a working default would build it, so the architecture can be reviewed before any RTL exists. It is deliberately cheap to change — one file, no chapter skeleton yet.

29.2 2. Feature sketches

29.2.1 2.1 Core datapath — the blocking bet

- **F1 — Blocking single-outstanding pipeline.** A miss freezes the front end until the fill completes and the original request replays. No MSHR, no hit-under-miss, no merge logic — that deletion is the entire point; it is jet's delta (jet J1–J5).
- **F2 — Geometry as elaboration parameters** (D1 OPEN). PROPOSED default profile 32 KiB / 64 B lines / 4 ways (128 sets) as the DV matrix centre of gravity; ranges 4–32 KiB, 32–64 B, 2–8 ways.

- **F3 — Pluggable replacement** (D7 OPEN). PROPOSED: true-LRU as the correctness reference (the cache simulator models it exactly), tree-PLRU as the FPGA-area variant; FIFO/random kept because the sim models them too. Policy is an elaboration parameter, never a code hack, so the sim↔RTL cross-check holds for every policy both sides implement.
- **F4 — Write policy** (D5 OPEN). PROPOSED: write-back + write-allocate — MESI's M state earns its keep, and dirty-victim handling is where blocking simplicity pays. Write-through/no-allocate retained as a bring-up mode because it deletes the victim path entirely.
- **F5 — Whole-line fills.** One burst per fill; sub-line fill granularity is a non-goal. Burst length derives from `LINE_BYTES / (BUS_WIDTH/8)`.

29.2.2 2.2 Coherence

- **F6 — MESI state per line** (D6 OPEN). PROPOSED plain MESI; MOESI's O-state recorded as a later elaboration upgrade, not v1 — ownership's win (dirty-transfer reduction between caches) is a measurement for the pair rig, not a v1 requirement. Insurance: allocate a 3-bit state field now (MESI needs 2; MOESI needs 3) so the upgrade is an encoding change, not an array rebuild.
- **F7 — ACE snoop responder** (D4 DECIDED — sketched, not re-decided). `amber_snoop_resp`: a thin adapter speaking raw ACE AC/CR/CD on the outside per the onyx D7 contract (IHI 0022 the definition) and driving a bus-agnostic probe interface inside. The cache core never contains ACE — only the adapter does; that is exactly why D4 was chosen (onyx, or a real ACE interconnect, can supersede the custom bus without touching cache RTL).
- **F8 — Snoop/local arbitration — the culsans problem.** Probes must not starve and CPU tag latency must stay bounded. PROPOSED: tag array on the dual-port `sdpram_core` — port A for CPU lookups, port B for snoop lookups; data array port A for CPU/fill, port B for snoop readout feeding
  CD. No arbitration stall in the common case; the *control* collision (miss FSM vs snoop FSM wanting the same array control port) resolved with snoop-priority round-robin inside `amber_control`.
- **F9 — Probe-during-fill, answered by a bypass** (direction set 2026-10-06: add the bypass). Blocking means a fill is usually in flight, and a probe against a pending line must not observe pre-fill state. The answer is not an unbounded stall: amber keeps one pending-fill register — single

outstanding by construction — recording {line address, state being installed, data availability}. A probe matching it is answered from the register: CRRESP reflects the post-fill state, and required CD beats are forwarded from the fill stream as its beats arrive. A probe that misses it takes the normal tag path. The snoop response is bounded either way. The correctness target sharpens rather than relaxes: *the bypass must be state-accurate* — a probe against a pending fill observes exactly the post-fill state, never pre-fill — alongside the response-bound liveness target. jet reopens the generalisation (jet J4/J6): multiple outstanding fills turn the register into a pending-fill table, which is jet's problem, not amber's.

- **F10** — **CRRESP response matrix** per MESI state, mirroring the RDS-DV ACE BFM default matrix as corrected in cocotb-framework 1.2.0 (MakeInvalid on M/O invalidates without data transfer — the bug the release fixed). The matrix ships as an explicit table in the full HAS.

29.2.3 2.3 Fabric side — where the two deployment rigs meet

- **F11** — **The two rigs.** *Pair rig* (amber D10): two ambers + shared memory, snoops peer-to-peer on the D4 port, misses are plain AXI4 to memory. *Onyx rig*: misses become coherent transactions toward onyx; onyx owns the memory side and drives the snoops. One cache core, two fabric attachments.
- **F12** — **Fabric side: two tops on one core** (Q1 answered 2026-10-06: do both, as two tops). The rig-agnostic cache core is shared; the fabric difference is confined to the top wrapper and its master stack:
  - `amber` — the pair-rig top. Misses are plain AXI4 to memory on the house `axi4_master_rd/wr` wrappers; snoop encodings never driven; the ACE snoop-responder port faces the peer cache.
  - `amber_ace` — the onyx-rig top. Fabric side is the existing `rtl/amba/ace/axi4ace_master_rd` / `axi4ace_master_wr` modules (ACE fields and RACK/WACK auto-pulse already in place), driven by a thin `amber_ace_issue` coherent-transaction issuer mapping cache events to the onyx D2 subset — read miss → ReadShared / ReadUnique (by desired exclusivity), write promotion → CleanUnique / MakeUnique, dirty eviction → WriteBack, clean eviction → Evict. Memory masters not exposed; the snoop-responder port faces onyx. No parameter selects between them: the rig is whichever top is instantiated. Both tops are compiled, and both are

exercised in the DV matrix — `amber` against a peer + memory environment, `amber_ace` against an onyx environment (§5).

- **F13 — Fill/drain engines** (D3 open at sketch time; decided PRD v0.6, 2026-10-07: house wrappers). Fill = one AR burst, whole line. Drain = a small victim buffer (PROPOSED depth 1)
  - one AW burst. Single outstanding by construction — no reorder buffers.

29.2.4 2.4 Observation (D8 DECIDED — sketched, not re-decided)

- **F14 — `amber_monlite` wrapper.** MonBus event set: hit (read/write), miss with class split (cold/capacity/conflict, when the golden model supplies the class), snoop (type, hit/miss, response class), eviction (clean/dirty), MESI state transitions, fill/drain start+end. Tallied by `monbus_tally_axil` exactly like every other component; the wrapper never stalls the observed port — drop-and-count. The observer's own gate cost is measured present vs absent (F18), so "the observer does not perturb what it measures" is a quantified claim, not an assertion.

29.2.5 2.5 Verification & formal surface (D9 OPEN — direction proposed)

- **F15 — Pattern-B cocotb grids.** GATE/FUNC/FULL over the geometry × policy × rig matrix. Instruments already exist: the RDS-DV `AXI4ACESnoopMaster` BFM drives amber's snoop port and `AXI4ACEMasterRead/ Write` drive the fabric side (cocotb-framework 1.2.0).
- **F16 — Golden-model parity.** Trace replay from `bin/apps/cache_sim/` at identical geometry + policy; hit/miss/miss-class counts must match exactly (PRD success criterion 1). The replacement policy parameter (F3) is what keeps this check meaningful across policies.
- **F17 — SymbiYosys targets** (PRD criterion 2): no stale data served after an external write (F9 ordering); no deadlock/livelock under fair arbitration. Proof surfaces named now so the RTL is built proof-shaped: `amber_control` FSM, `amber_snoop_resp`, tag-array port sharing, victim buffer handoff.

29.2.6 2.6 Cost & characterisation (PRD criterion 4)

- **F18 — Synthesis sweep.** Geometry params × replacement policy on Nexys A7 and/or Genesys 2; LUT/FF/RAM/fmax recorded in `docs/` the way pumice characterised. Includes the F14 observer-present vs absent delta.

29.3 3. Proposed block architecture

Proposed module set (`amber_*` prefix, per the cache-ip convention):

Module	Role	PRD anchor
`amber`	Pair-rig top: `amber_core` + plain AXI4 master stack + snoop responder toward the peer	D10
`amber_ace`	Onyx-rig top: `amber_core` + ACE master stack + coherent-transaction issuer + snoop responder toward onyx	D10, F12
`amber_core`	Shared rig-agnostic core, instantiated by both tops	—
`amber_cpu_frontend`	GAXI slave (D2 decided); request accept, response return, replay latch	D2
`amber_control`	Blocking pipeline FSM: lookup → hit / miss → evict → fill → replay; owns tag-array control arbitration (F8) and the pending-fill bypass register (F9)	D1, D5
`amber_tag_array`	`sdpram_core` tag + state store (3-bit state field, F6)	D11
`amber_data_array`	`sdpram_core` data store	D11
`amber_repl`	Pluggable replacement (policy elaboration parameter)	D7
`amber_victim`	Dirty victim buffer + drain trigger	D5
`amber_fill` / `amber_drain`	AXI4 read/write engines; plain `axi4_master_rd/wr`	D3

Module	Role	PRD anchor
	under `amber`, `axi4ace_master_rd/wr` under `amber_ace`	
`amber_snoop_resp`	ACE AC/CR/CD adapter; bus-agnostic inside — faces the peer in `amber`, onyx in `amber_ace`	D4
`amber_ace_issue`	Coherent-transaction issuer for the onyx D2 subset — `amber_ace` only	F12
`amber_monlite`	Observation wrapper (D8)	D8

Miss data flow: frontend accept → tag lookup (port A) → miss → `amber_repl` picks victim → dirty? `amber_victim` + drain engine WB → fill engine AR burst → data array write + state install → original request replays from the frontend latch → response.

Snoop data flow: AC arrives at `amber_snoop_resp` → tag lookup (port B) → line state determines CRRESP and whether CD beats follow → response driven. Conflict with an in-flight fill resolves per F9.

Observation taps sit on control-FSM decision points, not the datapath — the F8 port split means the datapath need not be touched to observe it.

29.4 4. Interface sketches (shapes, not pinouts)

Side	Sketch	Anchor
CPU-side	**DECIDED (PRD D2, 2026-10-06): GAXI slave** — house BFM/monitor coverage and skid/FIFO plumbing already exist, and STREAM can attach as first real consumer without an adapter. Must express single-beat reads, write-allocate fills, byte enables.	D2

Side	Sketch	Anchor
Fabric-side	Two tops, shared core (F12): `amber` = plain AXI4 masters to memory + AC/CR/CD snoop responder toward the peer; `amber_ace` = `axi4ace_master_rd/wr` + `amber_ace_issue` toward the CCU (onyx D2 subset: ReadShared / ReadUnique / CleanUnique / MakeUnique / WriteBack / Evict) + the same snoop responder toward onyx. The rig is the top you instantiate — no mode parameter.	D3, D4
Observation	MonBus through `amber_monlite`; no other debug surface.	D8
Register block	**None.** Parameter-only IP; all visibility through MonBus tallies, unlike every other component's APB CSR. Recorded explicitly so it is a decision and not an accident.	—

29.5 5. Parameter sketch (illustrative — feeds the DV matrix)

Parameter	PROPOSED default	Range / set
`SETS`	128	16–512
`WAYS`	4	2–8

Parameter	PROPOSED default	Range / set
LINE_BYTES	64	32–64
REPL_POLICY	lru	lru, tree_plru, fifo, random
WRITE_POLICY	wb_wa	wb_wa, wt_na
VICTIM_DEPTH	1	1–2
BUS_WIDTH	64	32–128

Rig is **not** a parameter: it is the top you instantiate — amber or amber_ace (F12). Derived: tag width = ADDR_WIDTH – $clog2(SETS) – $clog2(LINE_BYTES) + 3 (state field, F6); fill beats = LINE_BYTES / (BUS_WIDTH/8).

29.6 6. Open questions this sketch surfaces

#	Question	Where it resolves
Q1	~~One mode parameter amber_fabric vs separate adapters per rig~~ — **RESOLVED 2026-10-06:** two tops on a shared amber_core — amber (pair rig) and amber_ace (onyx rig); no rig parameter, both compiled, both in the DV matrix (F12)	Full HAS implements it
Q2	~~GAXI vs plain valid/ready CPU-side~~ — **RESOLVED by PRD D2 (2026-10-06, Sean): GAXI slave**	amber_cpu_frontend RTL
Q3	Geometry default vs Nexys A7 RAM budget (F2)	With the F18 cost sweep
Q4	3-bit state field headroom for MOESI (F6)	Cheap now, expensive later — recommend taking the insurance

#	Question	Where it resolves
Q5	~~Probe-during-fill stall-bound~~ — **RESOLVED 2026-10-06: add the pending-fill bypass** (F9); formal targets become bypass state-accuracy + response bound	Full HAS implements it; PRD criterion 2 depends on it
Q6	Replacement default: LRU reference vs tree-PLRU area (F3)	D7 closure, informed by F18

29.7 7. What the full HAS will add

The chapter book per the stream/andesite convention: front matter; purpose, conventions, definitions; use cases / key features / system context; graphviz block diagram; data flow; pin-level interfaces; performance targets; integration (clocking, reset, verification). The MAS follows when micro-architecture decisions land (array banking, pipeline staging). This Pre-HAS is the draft those books grow from — sections 2–6 above map onto chapters 2–5.

29.8 8. Cross-references

- amber PRD — binding requirements; decision record
- onyx PRD D7/D2 — snoop-port contract and coherent-transaction subset the ONYX rig speaks
- jet PRD — J-rows hold amber constant; nothing here may silently pre-decide jet (F9's stall policy is amber's; jet J4/J6 explicitly reopen it)
- References/AMBA_ACE_Interface_Definition.md and References/gem5-ruby-protocols/ — protocol definition and golden-protocol source
- RDS-DV cocotb-framework 1.2.0 ACE BFMs — the DV instruments F15 assumes

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Last Updated: 2026-10-06 **Maintained By:** amber architecture (pre-decision working document)

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30 jet — Product Requirements (DRAFT)

Version: 0.1 (draft, 2026-10-04) **Status:** scaffold only — inherits every DECIDED row of amber's PRD. Decided inherited rows so far: D8 observation (`*_monlite`, 2026-10-04) and D4 snoop transport (ACE-shaped AC/CD/CR per the onyx D7 contract, 2026-10-05). The rows below are the deltas jet must decide for itself; all are OPEN until they record a name and a date.

30.1 1. Purpose

The lockup-free (non-blocking) MESI snoopy L1 cache: amber with MSHRs. The research question is measurement — with coherence, arrays, and observation held constant, what do outstanding misses, hit-under-miss, and miss merging actually buy in latency and bandwidth, in simulation and on board?

30.2 2. Decisions jet adds to amber's

#	Decision	Candidates	What it drives
J1	MSHR organization	OPEN: full associativity keyed by address vs direct-mapped by line index vs CAM of pending lines; capacity 2–16 entries as an elaboration parameter	the miss queue, merge logic, formal surface
J2	Hit-under-miss scope	OPEN: hits to any resident line (default) vs hits only to lines not conflicting with a pending fill vs reads-only (writes still block)	hazard logic between fill port and hit port
J3	Miss merging policy	OPEN: merge only exact-line aliases (default) vs merge line + offset range	memory traffic, MSHR occupancy, fairness

#	Decision	Candidates	What it drives
		vs no merging (issue duplicate fetches, dedupe on return)	
J4	Ordering guarantees	OPEN: CPU-side responses in-issue-order vs out-of-order with ID tags; snoop-during-miss handling (a probe against a pending line must see the post-fill state or stall cleanly)	response path complexity, protocol proofs, consumer expectations
J5	Eviction under misses	OPEN: a dirty victim behind an outstanding fill must write back without deadlocking against the fill; victim buffer depth (0 = block like amber, 1–4 = decouple)	write-back path, the classic lockup-free deadlock corner
J6	Congestion/ fairness	OPEN: policy when MSHRs are full — stall new misses (strict) vs drop-and-retry vs throttle CPU-side accepts; plus snoop-vs-fill arbitration	starvation/ liveness proofs

#	Decision	Candidates	What it drives
		priority	
J7	The measured claim	OPEN: define the experiment now — same trace suite + same board as amber, report hit latency distribution, miss concurrency, bandwidth, and MSHR occupancy from MonBus capture	what “jet is faster by X” must mean, reproducibly

30.3 3. Success criteria (additions to amber’s)

1. **Correctness inheritance.** Every amber proof and golden-model parity result still passes with MSHRs present (proofs are re-run, not assumed).
2. **The delta, measured.** On the amber trace suite and at least one board: reported latency/bandwidth improvement with MonBus evidence, and the cache_sim cross-check extended so the model predicts jet’s concurrency numbers too (or the model gap is documented as a finding).
3. **The deadlock corner, proven.** J5/J6 formal targets: no deadlock when MSHRs are full and dirty victims wait behind fills.

30.4 4. Non-goals (for jet, beyond amber’s)

Speculative prefetching, adaptive replacement, quality-of-service policies, multi-banked parallelism beyond what MSHRs imply. Each is a future IP or a later decision.

31 onyx — Product Requirements (DRAFT)

Version: 0.5 (draft, 2026-10-11) **Status:** decisions pending — this page records the questions and the candidates, not answers. A decision becomes DECIDED when it records a name and a date, the way the reed-solomon PRD does it. Until then every row below is OPEN. **v0.5 records two owner rulings (N1 and invariant 6); all other D/N rows and invariants remain OPEN/PROPOSED.** v0.2: D7 decided 2026-10-05 (Sean) — onyx owns the ACE-shaped AC/CD/CR snoop port; amber/jet adapt to it behind a thin adapter. Decided together with amber D4, so the gate between the two is resolved. The sequencing constraint that remains: onyx still waits behind amber's pair rig (amber D10), its first consumer. v0.3: D10 direction tightened 2026-10-05 — the ACE BFM extension of the RDS-DV `axi4` family is now specified concretely (layering, four classes, build order); the row stays OPEN. RDS-DV implementation filed as RTLDesignSherpa-DV issue #86. v0.4: Chinnakonda book analysis folded in 2026-10-11 — Ch. 6/7/8/10/11/13/14 mined for candidate invariants, liveness scenarios, and scope-growth options. Book is cited in References; all new material is candidate/recommended/PROPOSED. v0.5: Ruling 2026-10-11 (Sean) — N1 decided (RACK/WACK omitted from the onyx↔️cache contract) and invariant 6 decided (CD-before-CR is the binding family convention). See also the updated AMBA ACE Interface Definition.

31.1 1. Purpose

A snoop-based cache coherency unit (CCU) in the shape of Arm ACE: the manager that sits between the repo's coherent caches and memory on the house AXI4 fabric. An attached cache (amber instance, jet instance) is an ACE *master*; it issues coherent transactions (ReadShared, ReadUnique, CleanUnique, MakeUnique, WriteBack, Evict) instead of managing peers directly. onyx receives those transactions, broadcasts a snoop (AC request) to the other caches, gathers their CR responses (DataTransfer / IsShared / PassDirty) and CD data, decides whether the transaction may be completed from cache-supplied data alone or memory must be read, and serialises same-address traffic. The research question is the manager side itself: what do fanout, gather, adjudication, and ordering cost — in gates, in latency, and in verification effort.

31.2 2. Background

ACE is AXI4 plus three snoop channels (AC, CR, CD) — and this repo already speaks AXI4 (skid buffers, GAXI FIFOs, `axi4_master_rd/wr` wrappers, Pattern-B BFMs). The natural-protocol argument: every piece of onyx's data path reuses existing, proven plumbing; the new logic is the coherence decision layer and the serializer. The definition is Arm IHI 0022 (the ACE chapters); the worked example is `pulp-platform/culsans` (ETH/PlanV), whose two halves are exactly the reading list:

- **culsans' CVA6 coherent WB cache** — how a cache becomes a snoop responder: the AC/CD/CR channels, how snoop tag lookups arbitrate with the core's own tag port, and the MOESI-ish line states.

- **`ace_ccu_top`** — the manager: broadcast snoops on a coherent read/write, collect responses, decide whether memory is read, and serialise same-address transactions.

Treat culsans as a worked example, not a dependency (last push 2024, lightly maintained). Note also what onyx is *not*: it is not a directory, not an L2, and not a general NoC — amber's deferred "snoop filter / directory-lite" question (amber D4) shows up here only as one candidate row (D3).

Repo anchors the design must respect: resets on the `ALWAYS_FF_RST` macros, no bespoke SRAM (shared `sdpram_core` / house FIFO primitives), `[DEPTH]` array syntax, valid/ready streaming contracts, MonBus observation through `*_monlite` wrappers only (amber D8's decision — the observer must never stall the measured paths), filelists registry, and GLOBAL_REQUIREMENTS.

31.3 3. Decisions that pick the design

#	Decision	Candidates	What it drives
D1	Functional scope	OPEN: CCU only — N ACE slave-side ports toward caches, one AXI4 master toward memory, one ACE-Lite IO port (the culsans shape; recommended starting scope) vs CCU + address striping / multiple memory ports (interconnect territory; non-goal today)**Book input:** §10.1 (banks ≥ master count), §10.2–10.3 (centralized vs. distributed manager; BIU as mapper), §10.7 (coherency logic	block boundary, what "onyx done" means

#	Decision	Candidates	What it drives
		can live in L2 or BIU). Confirms onyx as the BIU/manager block rather than an L2; keeps CCU-only the recommended v1 scope.	
D2	ACE transaction subset	OPEN: research subset — ReadShared / ReadUnique / CleanUnique / MakeUnique / WriteBack / Evict (recommended) vs subset + barrier support vs full ACE incl. DVM**Book input:** §7.2–7.4 give the full ARSNOOP/AWSNOOP taxonomy; WLU/WU are entire/partial-line writes used for non-temporal/all-bytes-updated stores and coherent mem-copy (§11.1); WC/WE are clean-victim/LLC-victim writes. The	front-end decoder, BFM scope, conformance matrix

#	Decision	Candidates	What it drives
		recommended research subset stays RS/RU/CU/MU/WB/Ev; WLU/WU/WC/WE are recorded as scope-growth option N5.	
D3	Snoop fanout	OPEN: broadcast to all attached caches (culsans shape, simplest, recommended for v1) vs directory-lite snoop filter as an elaboration parameter (amber D4's deferred question, reopened here) vs snoop filter mandatory**Book input:** §10.3–10.5 describe inclusive-directory options: copy-of-L1-tags, L1ID sharer vector in tag, or separate snoop filter sized ≈1.5× combined L1. §10.6.2 notes the false-positive-snoop corner	fanout logic, gate count, the coherence-traffic measurements

#	Decision	Candidates	What it drives
		(multiple shared responders → pick one; many systems read DRAM for shared copies). Broadcast remains recommended v1; directory-lite is a future elaboration parameter.	
D4	Response gather	OPEN: fixed response window (all caches respond within N beats, presumed-miss after timeout — simple, needs a proven bound) vs tagged outstanding snoop IDs with per-transaction completion (exact, more bookkeeping)**Book input:** §7.1 and §7.7 state snoops have no IDs and must be responded to in issue order; §14.18 confirms the three snoop channels are independent	gather buffer, formal surface, ACE conformance

#	Decision	Candidates	What it drives
		except for the same-address WB/Ev/WC/WE dependency. §7.7 gives the response bound: a snoop may be delayed only until the last data beat of a matching fill has arrived, otherwise it must respond not-present; holding it beyond that deadlocks.	
D5	Same-address serialization	OPEN: per-address scoreboard allowing interleaving of different addresses (recommended; the interesting manager) vs full in-order single outstanding (simplest, strangles concurrency; fine as a bring-up mode)**Book input:** §7.5, §10.7, §10.8 describe same-AxID ordering and the	the serializer, the deadlock/liveness proof, jet J4 interactions

#	Decision	Candidates	What it drives
		banked PoS with per-address linked lists (older completes first, different addresses any order); §10.9.1 notes device accesses routed to a dedicated bank for program order; §10.8 gives outstanding-access sizing math (≈40 read + 40 write transactions for 200-cycle DRAM latency on a 1 GHz 16-byte bus). Informs the scoreboard/serializer design but does not decide it.	
D6	Memory-read elision	OPEN: full adjudication — read memory only when no cache supplied data and no PassDirty (correct ACE semantics; recommended) vs always-read simplification for bring-up**Book input:** §10.5–	memory traffic, the measurable win onyx exists to quantify

#	Decision	Candidates	What it drives
		10.6.2 refine adjudication: when multiple DataTransfer responders exist, the manager must pick one; many real systems read DRAM for Shared copies and accept data only from an O-state holder; MOESIF Forward state is one way to designate a single forwarder. Therefore “any DataTransfer elides memory” is NOT the safe rule — elision requires a dirty (PassDirty or O-state) responder, or a single designated forwarder.	
D7	Cache-side interface ownership	**DECIDED 2026-10-05 (Sean): onyx defines and owns an AC/CD/CR-shaped snoop port; amber/jet stay bus-agnostic internally and add a thin**	the amber D4 ↔️ onyx contract, who changes when ACE supersedes the custom bus

#	Decision	Candidates	What it drives
		adapter to it. Chosen over caches natively growing ACE ports — the cache cores stay unpinned from ACE except at the adapter, so onyx (or a real ACE interconnect, e.g. culsans' `ace_ccu_top`) can supersede the custom bus without touching cache RTL.	
D8	Domain / barrier rules	OPEN: single shareability domain, barriers ignored (recommended non-goal, recorded here so it is a decision and not an accident) vs minimal barrier ordering vs full DVM**Book input:** §6.4–6.5 and §14.33 recommend making stores always multi-copy atomic even though Arm does	scope honesty, later upgrade path

#	Decision	Candidates	What it drives
		not require it; this is the book's justification for terminating barriers in the processor and ignoring them at the manager. Single shareability domain / barriers ignored remains recommended.	
D9	Observation	OPEN (direction: amber D8 pattern): `*_monlite` taps for coherent-txn start, snoop fanout per transaction, response gather completion, memory-read elision hit/miss, serialization stalls, and WB/Evict drains; tallied by `monbus_tally_axil`**Book input:** §8.2/§8.3/§10.6 treat independent snoop paths and fairness as first-class observables;	perf counters, board capture, DV scoreboard

#	Decision	Candidates	What it drives
		§10.8 gives outstanding-access sizing math that shapes the counters listed above.	
D10	Verification strategy	OPEN (direction: Pattern-B cocotb gate/func/full. **ACE BFM — an extension of the `axi4` BFM family in RDS-DV, layered the way that family is built** (analysis from inspecting `CocoTBFramework/components/axi4/`): (1) field configs — an `AXI4ACEFieldConfigHelper` adds `ARSNOOP[3:0]` / `AWSNOOP[2:0]` to the AR/AW field configs; (2) signal-map patterns — new `axi4ace_*` protocol patterns (decide there: framework-shipped vs. user-loadable, the signal_mapping_helper question);	DV structure, formal areas, CI time

#	Decision	Candidates	What it drives
		(3) transaction/packet layer — the six D2 transaction types as an enum, CRRESP as a bitfield (DataTransfer/Error/PassDirty/IsShared/WasUnique); (4) four BFM classes — `AXI4ACEMasterRead` / `AXI4ACEMasterWrite` as subclasses (snoop type on commands; onyx-port BFMs do not model RACK/WACK because the contract omits them), `AXI4ACESnoopSlave` (new, no base: receives AC, drives CR/CD in **issue order** — snoops carry no ID — with programmable cache-state behavior producing the correct CRRESP; this is the golden-	

#	Decision	Candidates	What it drives
		model hook that replays gem5 traces through amber), `AXI4ACESnoopMaster` (new: drives AC, gathers CR/CD in order — drives onyx's fanout in DV); (5) compliance checker — in-order CR/CD, no snoop to the requesting master, CD-before-CDLAST, CRRESP validity per snoop type; (6) main-repo TB classes under `bin/TBClasses/axi4ace/` per consumer. **Build order:** snoop-slave BFM first (amber's pair rig needs it), snoop-master BFM lands with onyx; ships as an RDS-DV framework release the way the irq BFM did. Python golden model derived	

#	Decision	Candidates	What it drives
		from the gem5 Ruby `MESI_Two_Level` spec in `../References/gem5-ruby-protocols/`; the ACE subset contract is the in-repo definition doc; SymbiYosys proofs of same-address serialisation and no-deadlock under fair arbitration; culsans behaviours as informal cross-checks, not oracles)**Book input:** §7.6–7.10, §8.1–8.3, §10.9 give concrete liveness scenarios: WLU-vs-WB valid/ready deadlock when SI buffers are full (§7.6.1/§7.7.1); CU livelock on RU-then-downgrade (§7.8, §10.9); store-exclusive semaphore	

#	Decision	Candidates	What it drives
		livelock and POS-monitor recipe (§7.9–7.10); starvation from fixed priority → timeout fairness (§8.3). These become the named proof obligations under Success criterion 3.	
D11	Master count & profile	OPEN: N coherent cache masters 2–4 as an elaboration parameter (culsans uses 2–4 CVA6) vs fixed 2 for the research rig; line size / profile locked to amber's geometry vs independently parameterised**Book input:** §10.1 states banks should be ≥ master count (4 CPUs → at least 4 banks). Supports keeping N=2–4 as an elaboration parameter and sizing internal resources accordingly.	array dimensions, DV matrix, the measured system's realism

#	Decision	Candidates	What it drives
N1	RACK/WACK omission-by-contract	**DECIDED 2026-10-11 (Sean): follow the book — RACK/WACK are omitted from the onyx↔️cache contract.** The binding condition is that data-return and snoop paths are ordered to the coherency point, which the D7 port contract guarantees. The house `axi4ace_master_rd/wr` wrappers still auto-pulse RACK/WACK; that behavior is deprecated for family contract use, and its removal is tracked as amber RTL follow-up A5.	adapter logic, snoop/data ordering assumptions
N2	AxLock / PoS monitor for cross-master exclusives	OPEN: arm a POS monitor on AxLock for exclusive transactions: one master receives EXOKAY, competing CUs receive OKAY	exclusive-transaction handling, POS monitor logic

#	Decision	Candidates	What it drives
		(fail), and invalidation snoops are withheld from the winning master until its store completes. Book §7.9–7.10 and §14.36 describe this as the livelock-reduction recipe for store exclusives. Out of the initial D2 subset; recorded as a candidate upgrade. Planned future consumer: gyrfalcon-A (falcon suite design spec amended 2026-10-11) on a coherent onyx rig; the row stays OPEN until that rig is designed.	
N3	Snoop-response sourcing from in-flight fills	OPEN: a snoop matching an in-flight fill must respond not-present if no data beat has arrived, and may forward from the fill buffer only after	snoop responder, fill-buffer comparators, formal progress proof

#	Decision	Candidates	What it drives
		the last data beat has arrived; the response must never depend on an unrelated event. Book §7.7, §8.2, §10.6. Elevates the D4 book input to a proposed invariant.	
N4	CD-path independence	OPEN: never share the snoop-data return path (CD) with WDATA; reserve credit/capacity for the WB/Ev/WC/WE family so snoop responses cannot be blocked by ordinary writes. Book §10.6 and §14.11–14.13 emphasize an independent snoop path as the deadlock-avoidance bedrock. (ACE permits CDDATA to be combined with WDATA; the book argues against it.)	snoop data channel, write-buffer allocation, credit logic

#	Decision	Candidates	What it drives
N5	WLU/WU + WC/WE scope-growth option	OPEN: add WriteLineUnique / WriteUnique for coherent DMA/mem-copy, plus WriteClean / WriteEvict for clean-victim handling. Book §7.3–7.4 and §11.1–11.3 tie WLU/WU to mem-copy traffic (read stream → RS+Evict, write stream → WLU/WU); WC/WE handle clean victims and LLC victim caching. Not in the recommended D2 subset; recorded as a scope-growth candidate tied to a future coherent DMA/mem-copy use-case.	front-end decoder, fanout/adjudication, conformance matrix
N6	DVM upgrade path	OPEN: if DVM is ever promoted, follow the book's recipe: DVM Messages are two-beat read transactions sent	DVM message handling, serialization, drain logic

#	Decision	Candidates	What it drives
		back-to-back with no intervening transaction; one DVM Sync per cluster is serialized at the SI; DVM Complete uses a unique ARID and is sent only after all translated-address buffers have drained. Book §7.16 (TLB maintenance / DVM). Revisits the D8/DVM non-goal but does not promote it.	
N7	Prefetch-traffic handling constraint	OPEN: the CCU must never see speculative store-ownership requests; prefetch traffic, if it reaches onyx at all, is speculative read-shared/read-clean only. Book §13.3: processors do not prefetch for stores because prefetches are speculative and must not make speculative state	prefetch filtering, BFM scenario matrix, coherency checker

#	Decision	Candidates	What it drives
		changes or set Dirty bits.	

31.4 4. Design invariants (proposed)

The following bedrock liveness rules are mined from Chinnakonda Ch. 7/8/10/14. Each is **PROPOSED** pending an owner ruling except where marked **DECIDED**.

1. **Snoop forward progress (PROPOSED).** A snoop may wait only on (a) a same-address outstanding WB/Ev/WC/WE, or (b) a guaranteed-to-complete temporary block (e.g. a retired store completing after it has received ownership). It must never wait on an unrelated transaction or an unpredictable buffer-drain event.
2. **Snoops ignore WLU/WU entirely (PROPOSED).** §7.13: a snoop to the same address as a WLU/WU is always allowed to make progress; the WLU/WU does not block it.
3. **Snoop responses never depend on WU/WLU completion (PROPOSED).** Otherwise SI buffers full of interdependent WU/WLU traffic can deadlock.
4. **CD never blocks on WDATA (PROPOSED).** The snoop-data return path must have independent forward progress; if it shares resources with ordinary writes, those resources must be credit-reserved for the WB/Ev/WC/WE family.
5. **Multi-copy-atomic stores (PROPOSED).** A store is considered complete only after all ownership invalidations have been performed everywhere; this is the book's recommended simplification that makes barriers-ignored at the manager defensible.
6. **CD before CR, CD-before-CDLAST, in-order CR/CD (DECIDED 2026-10-11 (Sean)).** A snoop transaction's CD beats complete before its CR is asserted; CR responses are returned in AC order; CD data is returned in AC order; CDLAST marks the final beat. This is the family contract, stricter than the spec's per-channel ordering rule.
7. **No snoop to the requesting master (PROPOSED).** Already a D10 compliance checker rule; restated here as a design invariant.

31.5 5. Success criteria

1. **Three-block coherence.** Two amber instances (or amber + jet) plus onyx form a coherent system that passes the golden-model parity suite amber D9 defines — the pair rig grown by one block, same traces, same policy space.
2. **IHI 0022 conformance on the D2 subset.** A scoreboard checks the AC/CR/CD behaviour per transaction type: the CR bits (DataTransfer, IsShared, PassDirty) of every gather are correct for the transaction and the caches' states.
3. **The serializer, proven.** Formal: no two same-address coherent transactions proceed incompatibly; no deadlock under fair arbitration at N ∈ {2, 3, 4} masters; the D4 response-gather mechanism cannot livelock (a presumed-miss bound exists and is proven). The Chinnakonda book supplies the named proof obligations: absence of CU livelock at the PoS, no WLU/WB-style deadlock in the fanout/gather path, and fairness (no fixed-priority starvation) under N ∈ {2, 3, 4}.
4. **Observation.** Every D9 event class visible on MonBus in simulation and captured on at least one FPGA board.
5. **Cost.** Synthesis/implementation numbers (LUT/FF/RAM, fmax) on Nexys A7 and/or Genesys 2, recorded in `docs/` the way pumice characterised.

31.6 6. Non-goals

Cache storage in onyx (it is a manager, not an L2 — tempting because a CCU's gather point is the natural place to cache; explicitly out), directory-based coherence (the D3 snoop-filter candidate is the closest this IP comes), ACE DVM and full barrier semantics (D8; revisited by candidate N6 as a future upgrade path, not promoted out of non-goals), cache stash/dealloc features, multi-CCU / multi-domain systems, and any CPU-side or upstream-master port beyond ACE-Lite IO. Each is a future IP or a future decision, not an onyx requirement.

31.7 7. Where the block sits

The fabric's coherence point, on three kinds of port: **cache-side**, N ACE-facing ports accepting coherent transactions from amber/jet instances acting as ACE masters (D7, decided 2026-10-05: onyx owns this port's ACE shape; caches adapt); **memory-side**, one AXI4 master toward the house memory path (D6 fixes when it is read); **inter-cache**, the snoop fanout — AC requests driven into each peer cache, CD snoop data and CR snoop responses returned. No CPU port, no register block beyond what observation needs (D9 tallies live in `monbus_tally_axil` like every other component). Standalone-testable with BFM caches long before amber exists: culsans' `ace_ccu_top` demonstrates exactly that decoupling.

31.8 8. References

1. Arm, *AMBA AXI and ACE Protocol Specification*, IHI 0022 (ACE chapters) — the normative definition. The in-repo working subset is recorded in `../References/AMBA_ACE_Interface_Definition.md`.
2. Murali Chinnakonda, *Designing a Multiprocessor Cache Subsystem* — Arm-ACE-centric multiprocessor cache subsystem design; ch. 7/8/10/14 mined 2026-10-11. (Source markdown in the owner's calibre library; not mirrored in-tree.)
3. `pulp-platform/culsans` — `rtl/src/ace_ccu_top.sv` and the CVA6 coherent cache: a working ACE CCU and a working snoop-responder cache. Worked example, not a dependency. https://github.com/pulp-platform/culsans
4. *A Primer on Memory Consistency and Cache Coherence* (Sorin, Hill, Wood, 2nd ed., stored alongside the ACE definition doc) — coherence-theory background the transaction types instantiate.
5. gem5 Ruby SLICC specs (`gem5-ruby-protocols/`, stored alongside the ACE definition doc) — executable transient-state tables for the same protocol family; the golden-model source material, phrased in messages rather than ACE signals.
6. amber/jet improvement plan memo — future family-level planning document at `../docs/amber-jet-improvement-plan.md`; linked for cross-reference even though it does not yet exist.

32 AMBA ACE — Working Interface Definition for cache-ip

Distilled 2026-10-05 for the `cache-ip` family, then **verified clause-by-clause against IHI 0022H on 2026-10-05** (Arm's direct PDF, linked in §7; fetched to `/tmp` for comparison, not stored in-tree). Verification confirmed every transaction type, channel payload, CRRESP bit, and the ACE-Lite channel exclusions; it caught and fixed two errors in the ordering rules (the original draft claimed ACE serializes snoops one-at-a-time and mandates CD-before-CR — ACE requires only per-channel ordering and permits multiple outstanding snoops; the stricter behavior is recorded above as our own adapter convention). This document is the in-repo definition of the **ACE-shaped port contract** referenced by amber D4 and onyx D7 (family decision 2026-10-05: cache-ip is ACE-shaped). Rulings recorded 2026-10-11 (Sean): RACK/WACK are omitted from the family contract because the D7 port guarantees ordered data-return and snoop paths; CD-before-CR on the snoop response remains the family convention, confirmed against the spec's per-channel ordering rule.

Normative reference: Arm, *AMBA AXI and ACE Protocol Specification*, IHI 0022 (ACE chapters). Available from Arm without charge (direct PDF in §7); Arm copyright, so deliberately not mirrored here. Where this document and IHI 0022 disagree, IHI 0022 wins — file a docs bug.
Worked example: `pulp-platform/culsans` (`ace_ccu_top`, Solderpad license) — read its RTL after this page, not before.

32.1 1. What ACE is

ACE (AXI Coherency Extensions) is AXI4 extended so that multiple caching masters can share memory coherently. An ACE master is typically a CPU's cache; an ACE-Lite master is a non-caching agent (DMA, IO) that wants coherent *access* without being snooped. Between the masters and memory sits a coherency manager — in this family, **onyx** — which receives coherent transactions, broadcasts snoops to the peer caches, gathers their responses, decides whether DRAM must be read, and serializes same-address traffic.

ACE adds three channels to AXI4. Everything else — AW/W/B/AR/R semantics, burst rules, handshake rules — is unchanged AXI4 and is not restated here.

32.2 2. Channel inventory

32.2.1 2.1 Coherent master → manager (front side, per master)

The five AXI4 channels, with a transaction-type field added to each address channel:

Channel	New/changed payload	Notes
AR + R	`ARSNOOP[3:0]`	read-side coherent

Channel	New/changed payload	Notes
		transaction type
AW + W + B	`AWSNOOP[2:0]`	write-side coherent transaction type

Data and response channels are plain AXI4. A coherent read (e.g. ReadShared) delivers its data on the ordinary R channel, sourced by the manager from a peer cache or from memory as the adjudication decides.

Full ACE master interfaces also carry two acknowledge handshakes outside the five channels: **RACK** (read acknowledge — master pulses after the last read data beat of each transaction) and **WACK** (write acknowledge — master pulses after the write response). ACE-Lite excludes both (IHI 0022H §D11.1). Ruling 2026-10-11 (Sean): the family contract omits RACK/WACK; per Chinnakonda §7.20/§14.39 they are unnecessary when data-return and snoop paths are ordered to the coherency point, which the D7 port contract guarantees; the house master wrappers still carry auto-pulsed RACK/WACK pins, deprecated for family use (removal: amber A5).

32.2.2 2.2 Manager → cache, cache → manager (snoop side, per cache)

Three new channels. Direction below is from the point of view of the cache (amber/jet); the manager (onyx) sees the mirror image.

Channel	Dir	Payload	Meaning
AC — snoop address	in	`ACADDR`, `ACSNOOP[3:0]`, `ACPROT[2:0]`	“What state does this line have in you, and what should you do about it?”
CR — snoop response	out	`CRRESP[4:0]`	the answer: will you transfer data, is the line shared, is dirty responsibility passing
CD — snoop data	out	`CDDATA`, `CDLAST`	the line’s data, when CRRESP said it would come

Handshake rules mirror AXI4 (VALID/READY, no combinatorial dependency of READY on VALID). Ordering rules that matter (IHI 0022H §D3.8–§D3.9, §D5.2, §D6.2):

- **CR and CD only after AC.** The snoop response and any snoop data must only be asserted after the ACVALID/ACREADY handshake.
- **In-order responses, in-order data.** CR responses must be returned in the same order as the AC addresses; CD data likewise. The spec permits a cache to have **multiple outstanding snoop transactions** — serialization is not required by ACE, only per-channel ordering is.
- **Family convention (decided with onyx D4):** the amber/jet adapter completes a transaction's CD beats before asserting its CR, so onyx's gather sees one fully-closed response at a time. This is stricter than ACE requires and simpler to verify; it is our contract, not the spec's. Confirmed by ruling 2026-10-11 (Sean) after the book yielded no guidance — the spec permits either order (per-channel ordering only, verified against IHI 0022H §D3.8–D3.9, §D6.2 on 2026-10-05); the family convention stands.
- **Snoop vs. local access arbitration.** A snoop reads the tag array and may evict/change a line's state, so it must arbitrate with the cache's own CPU-side tag port. This arbitration point is the classic cache-side cost of becoming a snoop responder (culsans' CVA6 cache shows one arrangement).

32.3 3. Transaction types (the onyx D2 research subset in bold)

Read side (`ARSNOOP`): ReadOnce, **ReadShared**, **ReadUnique**, ReadClean, ReadNotSharedDirty, **CleanUnique**, **MakeUnique**, plus barrier/DVM forms (out of scope, onyx D8).

Write side (`AWSNOOP`): **WriteUnique**, WriteLineUnique, **WriteBack**, WriteClean, **Evict**, plus barrier/DVM forms (out of scope, onyx D8).

The six bolded transactions are the subset onyx D2 recommends; the definition of the subset is: everything amber or jet needs to run MESI write-back coherence against a manager, and nothing that requires DVM or barrier semantics.

Intent of each subset member, one line each:

- **ReadShared** — fetch a line to read; other caches may keep Shared copies; the line arrives in S (or E if nobody shared it).
- **ReadUnique** — fetch a line to write; all other copies must be invalidated; if a peer held it dirty, its data (or DRAM's) is supplied.
- **CleanUnique** — "I already hold the line; make my copy Unique." No data transfer; peers with Shared copies drop them.

- **MakeUnique** — like CleanUnique but the requester does not need the data (it will overwrite the whole line); peers invalidate.
- **WriteBack** — retire a dirty line to memory; the manager must merge the data into DRAM (or hand it to a subsequent reader).
- **Evict** — retire a clean line; a hint: no data, no DRAM write needed.

32.4 4. The CRRESP decision bits

CRRESP is where a snooped cache answers. The four bits the manager's adjudication logic consumes (bit 4, `WasUnique`, also exists per IHI 0022):

Bit	Name	Meaning when set
[0]	DataTransfer	"I will drive the line's data on CD." Required of a Modified responder (or any responder told to transfer).
[1]	Error	"My lookup had an error." Propagated, not interpreted.
[2]	PassDirty	"The line is/was dirty and dirty responsibility moves to whoever receives the data" — set with DataTransfer when the responder passes dirty data, and on its own when the data is already moving elsewhere. The manager uses it to decide whether DRAM must be (re)written.
[3]	IsShared	"Another master (or I) may hold a copy." Tells the manager the requester must not be

Bit	Name	Meaning when set
		given Exclusive/Unique state, and that DRAM-read elision is safe when data was supplied.

The adjudication onyx D6 specifies falls straight out of these bits: a coherent read completes from cache-supplied data alone iff some CR carried `DataTransfer`; DRAM must be read iff none did and no `PassDirty` moved dirty responsibility; DRAM must be written iff `PassDirty` passed dirty data that the requester did not absorb.

32.5 5. ACE-Lite

ACE-Lite is the non-snooped half: a master may issue coherent transaction types but never receives AC, so it never drives CR/CD. onyx's IO port (onyx D1) is ACE-Lite — this is why the earlier "ACE-lite-shaped snoop channel" candidate in amber D4 was set aside: a snoop *responder* must drive CR/CD, and ACE-Lite has no snoop channels at all.

32.6 6. What this family implements, and what it deliberately does not

In scope (amber D4 / onyx D2 subset, single shareability domain, onyx D8): the six transaction types of §3, the AC/CR/CD contract of §2, the CRRESP semantics of §4, broadcast snoop fanout (onyx D3 v1), same-address serialization (onyx D5).

Transport for this contract exists in `rtl/amba/ace/` (added 2026-10-05): `axi4ace_master_rd/wr` and `axi4ace_slave_rd/wr` (front-side movers with `ARSNOOP`/`AWSNOOP`; auto-pulsed `RACK`/`WACK` on the house master wrappers is deprecated per ruling 2026-10-11), the `axi4ace_snoop_slave` / `axi4ace_snoop_master` AC/CR/CD movers, and `rtl/amba/ocelus/axi4ace_snoop_monitor_lite` for snoop-side observation.

Out of scope, recorded as decisions and not accidents (onyx D8): DVM (distributed virtual memory) messages, barrier ordering, multiple shareability domains, stash/dealloc hints. The repo's port contract is **ACE-shaped** — same channels, signals, payload fields, and ordering rules on the pins — without claiming whole-spec conformance; see onyx D7 for why "shaped" rather than "compliant."

32.7 7. Sources

1. Arm, *AMBA AXI and ACE Protocol Specification*, IHI 0022, ACE chapters — the definition. Current issue H (2020), direct PDF: https://developer.arm.com/-/media/Arm%20Developer%20Community/PDF/IHI0022H_amba_axi_protocol_spec.pdf (landing page:

https://developer.arm.com/documentation/ihi0022/). Available from Arm without charge; Arm copyright — cited, not mirrored. Sections this document was verified against: D2.2 (channels), D3.8–D3.9 (signaling, dependencies), D3.20–D3.22 (CRRESP), D5 (snoop transactions), D6.2 (sequencing), D11.1 (ACE-Lite).

2. `pulp-platform/culsans` — `rtl/src/ace_ccu_top.sv` and the CVA6 coherent cache: a working ACE CCU and a working snoop-responder cache. Worked example, not a dependency. https://github.com/pulp-platform/culsans
3. *A Primer on Memory Consistency and Cache Coherence* (Sorin, Hill, Wood, 2nd ed., stored alongside this file) — the coherence-theory background the transaction types above instantiate.
4. gem5 Ruby SLICC specs (`gem5-ruby-protocols/`, stored alongside) — executable transient-state tables for the same protocol family; the golden-model source material, phrased in messages rather than ACE signals.

33 gem5 Ruby protocol specs (reference extract)

Executable, state-table-style cache-coherence protocol specifications in gem5's SLICC language, extracted verbatim from gem5 `master` at commit `f5c5a6e390f55dd5984977815bf9d0bd05da6945` (2026-09-07) via sparse clone.

Two protocols, as requested for the cache-ip research library:

- **MESI_Two_Level** — classic snoopy MESI with a shared L2. The L1 cache controller (`MESI_Two_Level-L1cache.sm`) is the closest executable model to amber: every transient state (IS, IM, SM, S^I, M^I, …) between a stable MESI state and a pending response is enumerated as an explicit state with its own transition table. `MESI_Two_Level-msg.sm` is the message-type contract (the shape a D4 snoop transport must express); `MESI_Two_Level-dir.sm` + `-L2cache.sm` show the point where a passive memory-side agent has to make coherence-visible decisions.
- **MOESI_CMP_directory** — the directory-based contrast set: what changes when a home/directory agent (not a broadcast medium) owns coherence serialization. Useful for ruling D4 options in or out with a working executable spec instead of prose.

Why these exist here: the Primer's MESI diagrams show only stable states; most coherence bugs live in the transient states. These files are the enumerated transient-state tables — the raw material for deriving amber's `amber_pkg` state encoding and for building the Python reference model the DV scoreboard replays against (amber D9, jet J7 cross-checks).

License: each `.sm`/`.slicc` file carries its complete copyright and BSD-3-clause license header inline (ARM Limited; Mark D. Hill and David A. Wood) — redistribution terms are inside the files themselves. `LICENSE` is gem5's root file, an unrendered BSD-3 template, included as-shipped.

Upstream: https://github.com/gem5/gem5 (`src/mem/ruby/protocol/`). This directory is a read-only extract; track upstream for changes.

34 Cache Associativity Simulator

A standalone, trace-driven cache simulator for exploring set count, way count, block size, and replacement policy. Written in vanilla HTML5/CSS/ES5-ish JavaScript with no build step; open `index.html` directly as `file://`.

34.1 Files

```
bin/apps/cache_sim/
  index.html          -- UI
  style.css           -- mobile-first styles
  js/model.js         -- pure simulation model (no DOM)
  js/app.js           -- DOM glue, generators, and guided example
  test/run_tests.js   -- zero-dependency TAP-ish node test runner
  README.md           -- this file
```

34.2 Model API

```
var CACHESIM = require('./js/model.js');

var result = CACHESIM.simulate(
  { sets: 4, ways: 1, blockSize: 1, policy: 'LRU' },
  [0x00, 0x10, 0x20, 0x00],
  1   // seed for RANDOM policy / reproducibility
);
```

`simulate(config, addresses, seed)` returns:

```
{
  config: { sets, ways, blockSize, policy },
  perAccess: [
    { addr, hit, setIndex, way, missClass }, ...
  ],
  totals: {
    hits, misses, hitRate,
    compulsory, capacity, conflict
  },
  occupancy: [ [tagOrNull, ...], ... ]   // sets x ways
}
```

`missClass` is one of `'hit'`, `'compulsory'`, `'capacity'`, or `'conflict'`. Capacity classification is computed by running a shadow fully-associative cache with the same total size and policy.

Other model helpers:

- `CACHESIM.validateConfig(config)` -> `{ok, errors}`
- `CACHESIM.parseAddressText(text)` -> address array (NaN for bad lines)

- `CACHESIM.generateSequential(length, start)`
- `CACHESIM.generateStrided(length, start, stride)`
- `CACHESIM.generateRandom(length, seed, maxAddr)` (default maxAddr = 256)

34.3 Running tests

```
node bin/apps/cache_sim/test/run_tests.js
```

Exit code is 0 when all tests pass.

34.4 UI features

- Sliders for sets, ways, and block size; dropdown for policy.
- Address trace textarea plus one-click sequential, strided, and random generators.
- Big-number hit rate, totals table, per-access log, and per-set occupancy.
- Guided example button that loads a cyclic trace and compares direct-mapped vs 2-way set-associative behavior.

35 Global Hard Requirements for RTL Design Sherpa

Version: 1.0 **Created:** 2025-10-31 **Purpose:** Consolidated mandatory requirements extracted from all CLAUDE.md files

---

35.1 Overview

This document consolidates all **MANDATORY** requirements found across repository CLAUDE.md files. These are **HARD REQUIREMENTS** that must be followed - not suggestions or guidelines.

Sources Analyzed: - Root `/CLAUDE.md` - `projects/components/CLAUDE.md` - `projects/components/dma-ip/rapids/CLAUDE.md` (cited below as `rapids/CLAUDE.md`) - `projects/components/dma-ip/stream/CLAUDE.md` (cited below as `stream/CLAUDE.md`) - `rtl/amba/CLAUDE.md` (cited below as `amba/CLAUDE.md`) - `rtl/common/CLAUDE.md` - `CocoTBFramework/CLAUDE.md` - the DV framework guide, now maintained in the separate RTLDesignSherpa-DV repo (the framework is editable-installed into the venv)

35.2 Category 1: RTL Standards

35.2.1 1.1 Reset Handling (MANDATORY - NO EXCEPTIONS)

Requirement: ALL RTL in projects/components/ MUST use standardized reset macros

Pattern:

```
`include "reset_defs.svh"

`ALWAYS_FF_RST(clk, rst_n,
    if (`RST_ASSERTED(rst_n)) begin
        r_state <= IDLE;
    end else begin
        r_state <= w_next_state;
    end
)
```

NOT Allowed:

```
// WRONG - Manual reset handling
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        r_state <= IDLE;
    end else begin
        r_state <= w_next_state;
```

```
    end
end
```

Applies to: ALL synthesisable RTL – `rtl/` and `projects/` alike. **Source:** `projects/components/CLAUDE.md` Rule #0, widened 2026-09-06 (Sean: "All flops should use the global macro").

Reset is asynchronous on assertion. Always. `ALWAYS_FF_RST` used to be conditional on `USE_ASYNC_RESET` and defaulted to SYNCHRONOUS, so `make lint` (which set the define) and simulation/synthesis (which did not) disagreed about what the design was. The define is now a no-op; passing it is harmless and changes nothing. Deassertion still has to be synchronised externally.

The one exception, and it is structural: `rtl/cdc/reset_sync.sv`. It is active-HIGH, which the macro cannot express without `RESET_ACTIVE_HIGH` set globally, so it keeps a raw `always_ff`. Documented at the flop itself. If you add another, document it there too and say WHY – "it has always been like that" is not a reason.

"Macro-free fork" is not a reason. Two trees claimed it and neither survived contact with evidence. `timing_characterization/rtl/asic_only/` was a 37-file duplicate of `rtl/` differing ONLY in reset spelling; synthesised through Yosys + slang against ASAP7 it produced a byte-identical mapped netlist (396 cells, 97 flops, every cell type matching), and the flow reading it still passed `-DUSE_ASYNC_RESET`, a define meaningful only to the macro tree it was cloned from. `formal/cdc/cdc_handshake/cdc_handshake_formal.sv` claimed Yosys could not take the macro, while 92 other .sby units stage `reset_defs.svh` and read it with `-Iincludes`; its own .sby had simply never staged it, and it now proves PASS with the macro. The macros exist to serve ASIC and FPGA both – that is the whole point of them – so a second copy of a module to spell reset differently is duplication, not portability. Deleted 2026-09-07.

Status: `rtl/` converted 2026-09-06 (45 blocks / 20 files; the claim that it was "already compliant" was wrong). `projects/components/utility-ip/converters/` done as projects/components/utility-ip/converters BUG-007 (was CONV-009). Hand-written RTL still outstanding elsewhere under `projects/`, plus ~928 blocks in GENERATED files, which must be fixed at their generators and never in the emitted `.sv`.

35.2.2 1.2 FPGA Synthesis Attributes (MANDATORY)

Requirement: ALL memory arrays MUST have FPGA synthesis hints

Pattern:

```
`ifdef XILINX
    (* ram_style = "auto" *)
`elsif SYNTH_PRAGMA
    /* synthesis ramstyle = "AUTO" */
`endif
logic [DATA_WIDTH-1:0] mem [DEPTH];
```

Attributes: - Xilinx: `ram_style = "auto" | "block" | "distributed" | "ultra"` - Quartus: `ramstyle = "AUTO" | "M20K" | "MLAB" | "logic"`

Applies to: All memory array declarations **Source:** `projects/components/CLAUDE.md` Rule #1

35.2.3 1.3 Array Syntax Standard (MANDATORY)

Requirement: Use `[DEPTH]` NOT `[0:DEPTH-1]`

Pattern:

```
// CORRECT
logic [DATA_WIDTH-1:0] mem [DEPTH];

// WRONG
logic [DATA_WIDTH-1:0] mem [0:DEPTH-1];
```

Reason: Iverilog compatibility, cleaner syntax **Applies to:** All unpacked array declarations **Source:** `projects/components/CLAUDE.md` Rule #2

35.2.4 1.4 SRAM Module Standards (MANDATORY)

Requirement: SRAM modules DO NOT have reset ports

Pattern:

```
module simple_sram #(
    parameter int DATA_WIDTH = 32,
    parameter int DEPTH = 1024
) (
    input  logic clk,
    // NO rst_n port!
    // ... other ports
);
```

Reason: - Real SRAM chips don't have reset pins - FPGA block RAM doesn't reset contents - Initialize via writes during system init

Applies to: All SRAM/memory modules **Source:** `projects/components/CLAUDE.md` Rule #3

35.2.5 1.5 Reset Convention (STANDARD)

Requirement: ALWAYS use active-low asynchronous reset

Signal Names: - `rst_n` - Active-low reset (projects/components/) - `aresetn` - Active-low asynchronous reset (AMBA standard) - `i_rst_n` - Active-low reset (rtl/common/)

NOT Allowed: - `rst` (positive reset) - `reset` (ambiguous polarity)

Applies to: All RTL modules **Source:** Root `/CLAUDE.md`, `rtl/common/CLAUDE.md` Rule #3

35.3 Category 2: Testbench Architecture

35.3.1 2.1 TB Location Requirements (MANDATORY)

Requirement: Project-specific TB classes MUST be in project area, NOT framework

Directory Structure:

```
CORRECT:
projects/components/{name}/dv/tbclasses/{module}_tb.py

WRONG:
bin/TBClasses/{name}/{module}_tb.py
```

Import Pattern:

```
# Add repo root to path
import os, sys
repo_root = os.path.abspath(os.path.join(os.path.dirname(__file__),
'../../../../../..'))
sys.path.insert(0, repo_root)

# Import from PROJECT AREA
from projects.components.dma_ip.rapids.dv.tbclasses.scheduler_tb
import SchedulerTB

# Shared framework utilities
from TBClasses.shared.tbbase import TBBase
from CocoTBFramework.components.axi4.axi4_master import AXI4Master
```

Applies to: ALL project-specific testbenches **Source:** Root `/CLAUDE.md`, `projects/components/CLAUDE.md` Rule #4, `rapids/CLAUDE.md` Rule #0.1, `stream/CLAUDE.md` Rule #0.1, `amba/CLAUDE.md` Rule #0

35.3.2 2.2 Three Mandatory TB Methods (MANDATORY)

Requirement: EVERY testbench class MUST implement these three methods

Pattern:

```
class MyModuleTB(TBBase):
    async def setup_clocks_and_reset(self):
        """Complete initialization - starts clocks and performs
reset"""
        await self.start_clock('clk', freq=10, units='ns')

        # Set config before reset (if needed)
        self.dut.cfg_param.value = initial_value

        # Reset sequence
        await self.assert_reset()
        await self.wait_clocks('clk', 10)
        await self.deassert_reset()
        await self.wait_clocks('clk', 5)

    async def assert_reset(self):
        """Assert reset signal"""
        self.dut.rst_n.value = 0  # Active-low

    async def deassert_reset(self):
        """Deassert reset signal"""
        self.dut.rst_n.value = 1
```

Why Required: - Consistency across all testbenches - Reusability for mid-test resets - Clear test structure and intent - Configuration control (some RTL needs signals set BEFORE reset)

Applies to: ALL testbench classes **Source:** Root `/CLAUDE.md`, `rapids/CLAUDE.md` Rule #0.5, `stream/CLAUDE.md` Rule #0.2, `CocoTBFramework/CLAUDE.md` Rule #3

35.3.3 2.3 TBBase Inheritance (MANDATORY)

Requirement: ALL testbench classes MUST inherit from TBBase

Pattern:

```
from TBClasses.shared.tbbase import TBBase

class MyModuleTB(TBBase):
    def __init__(self, dut):
        super().__init__(dut)
        # Testbench-specific initialization
```

What TBBase Provides: - Clock management (`start_clock`, `wait_clocks`) - Reset utilities (`assert_reset`, `deassert_reset`) - Logging (`self.log`) - Progress tracking (`mark_progress`) - Common utilities (`convert_to_int`, `format_dec`)

Applies to: ALL testbench classes **Source:** `CocoTBFramework/CLAUDE.md` Rule #2

35.3.4 2.4 Three-Layer Architecture (MANDATORY)

Requirement: Verification MUST be separated into three layers

Layers:

1. **Testbench (TB):** `projects/components/{name}/dv/tbclasses/{module}_tb.py`
   - Infrastructure + BFMs
   - Clock/reset control
   - Base test methods
   - NO verification logic
2. **Scoreboard:** `bin/TBClasses/scoreboards/{protocol}/{module}_scoreboard.py`
   - Verification logic
   - Expected vs actual comparison
   - Coverage collection
3. **Test:** `projects/components/{name}/dv/tests/test_{module}.py`
   - Test intelligence
   - **Imports** TB class (does NOT define it)
   - Pytest wrappers
   - Scenario logic

NEVER Embed TB Classes in Test Files

Applies to: ALL verification code **Source:** `amba/CLAUDE.md` Rule #0, `rapids/CLAUDE.md` Rule #0.1, `CocoTBFramework/CLAUDE.md` Rule #4

35.3.5 2.5 Queue-Based Verification (PREFERRED)

Requirement: Use direct queue access for simple tests, NOT memory models

Pattern:

```
# CORRECT: Direct queue access
aw_pkt = self.aw_monitor._recvQ.popleft()
w_pkt = self.w_monitor._recvQ.popleft()

# Verify
assert aw_pkt.addr == expected_addr
assert w_pkt.data == expected_data

# WRONG: Memory model for simple test
memory_model = MemoryModel()
written_data = memory_model.read(addr, 4)  # Unnecessary complexity
```

When to Use: - **Queue Access:** Simple in-order tests, single-master, no overlap - **Memory Models:** Complex OOO scenarios, multi-master with address overlap

Applies to: Simple in-order verification scenarios **Source:** `rapids/CLAUDE.md` Scoreboard Pattern, `CocoTBFramework/CLAUDE.md` Rule #5

35.4 Category 3: Test Standards

35.4.1 3.1 Test Naming Convention (MANDATORY)

Requirement: Pytest function MUST match module name

Pattern:

```
# Module: axi4_dwidth_converter_wr.sv

CORRECT:
def test_axi4_dwidth_converter_wr(request, params):
    """Test for axi4_dwidth_converter_wr.sv"""
    ...

WRONG:
def test_converter(request, params):  # Too generic!
    ...
```

Why: Prevents pytest collection conflicts between related modules

Applies to: ALL pytest test functions **Source:** Root `/CLAUDE.md`

35.4.2 3.2 Test Pattern - CocoTB + Pytest (MANDATORY for projects/components/)

Requirement: Use two-tier pattern with cocotb_test_* prefix

Pattern:

```
# 1. CocoTB test functions - prefix with "cocotb_test_*"
@cocotb.test(timeout_time=100, timeout_unit="ms")
async def cocotb_test_basic(dut):  # ← cocotb_test_* prefix!
    """CocoTB test function - NOT collected by pytest"""
    tb = SimpleSRAMTB(dut)
    await tb.setup_clocks_and_reset()
    # ... test logic

# 2. Pytest wrapper - calls specific cocotb_test_* function
@pytest.mark.parametrize("addr_width, data_width", params)
def test_basic(request, addr_width, data_width):
    """Pytest wrapper - calls cocotb_test_basic"""
    run(
        # ... setup
        testcase="cocotb_test_basic",  # ← Explicitly specify
        # ... compilation args
    )
```

Why: - Pytest collects only wrappers, not CocoTB functions - Each wrapper calls ONE specific CocoTB function - Allows parameter sweeps at pytest level

Applies to: ALL tests in projects/components/ **Source:** Root `/CLAUDE.md`

35.4.3 3.3 100% Success Requirement (MANDATORY)

Requirement: ALL tests MUST achieve 100% success rate

Pattern:

```
# WRONG: Accepting partial success
assert success_rate >= 70  # "Good enough" - NO!

# CORRECT: Require 100% success
assert success_rate >= 100  # 100% required
```

Rationale: - RTL is deterministic - 100% success is achievable - Partial success indicates bugs or timing issues - Lower thresholds mask real problems - Tests must reliably catch regressions

Applies to: ALL tests **Source:** `rapids/CLAUDE.md`, `CocoTBFramework/CLAUDE.md`

35.4.4 3.4 Simulation Timestamps in Logs (MANDATORY)

Requirement: ALL test log messages MUST include simulation time

Pattern:

```
# CORRECT: Include simulation timestamp
self.log.info(f"@ {cocotb.utils.get_sim_time('ns')}ns: Beat {i}:
data=0x{data:X}, s_ready={ready}, occupancy={occ}")

# WRONG: No simulation timestamp
self.log.info(f"Beat {i}: data=0x{data:X}, s_ready={ready},
occupancy={occ}")
```

Output Example:

```
2025-11-06 19:11:20,688 - INFO - @ 150ns: Beat 1: data=0xA000,
s_ready=1, occupancy=0
2025-11-06 19:11:20,688 - INFO - @ 160ns: Beat 2: data=0xA001,
s_ready=1, occupancy=1
2025-11-06 19:11:20,688 - INFO - @ 170ns: Beat 3: data=0xA002,
s_ready=1, occupancy=2
2025-11-06 19:11:20,688 - INFO - @ 180ns: Beat 4 BLOCKED: s_ready=0,
occupancy=3
```

Why Critical: - Correlate test events with waveform timing - Debug timing-sensitive failures - Understand cycle-accurate behavior - Match log messages to specific clock edges

What Counts as "Simulation Time": - Use `cocotb.utils.get_sim_time('ns')` for nanoseconds - Use `cocotb.utils.get_sim_time('ps')` for picoseconds - Format: `@ {time} {unit}:` at START of message

Anti-Pattern:

```
# WRONG: Only Python logging timestamp
self.log.info(f"Beat {i} completed")
# Output: "2025-11-06 19:11:20,688 - INFO - Beat 1 completed"
# Problem: Can't correlate with waveform!
```

Applies to: ALL test log messages showing DUT behavior **Priority:** P0 - CRITICAL for debugging **Source:** User feedback 2025-11-06

35.5 Category 4: Framework Usage

35.5.1 4.1 Search Before Creating (MANDATORY)

Requirement: ALWAYS search existing modules/components BEFORE creating new

Search Commands:

```
# Search for existing RTL modules
find rtl/{subsystem}/ -name "*.sv" | xargs grep -l "keyword"

# Search for existing BFMs (CocoTBFramework, editable-installed from
RTLDesignSherpa-DV)
find "$(python3 -c 'import
CocoTBFramework,os;print(os.path.dirname(CocoTBFramework.__file__))')/
components" -name "*.py" | head

# Search for existing TB classes
find bin/TBClasses/ -name "*_tb.py"
```

Decision Tree: - Exact match exists → Use it - Close match exists → Adapt with parameters - No match found → Document search, propose new - Didn't search → STOP, go back and search

Applies to: RTL modules, BFMs, TB components **Source:** `rtl/common/CLAUDE.md` Rule #1, `CocoTBFramework/CLAUDE.md` Rule #1

35.5.2 4.2 BFM Usage for Protocols (MANDATORY for RAPIDS)

Requirement: MUST use framework BFMs for all protocol interfaces

BFM Selection: - **Custom valid/ready:** GAXI Master/Slave (`CocoTBFramework/components/gaxi/`) - **AXI4:** AXI4 Master/Slave (`CocoTBFramework/components/axi4/`) - **APB:** APB drivers (`CocoTBFramework/components/apb/`) - **AXI-Stream:** AXIS drivers (`CocoTBFramework/components/axis4/`)

(BFMs come from the RTLDesignSherpa-DV framework repo; shared TB classes in this repo are `bin/TBClasses/`)

NEVER: - Manually drive valid/ready handshakes - Create custom protocol drivers when framework has them

Applies to: All RAPIDS FUB-level tests **Source:** `rapids/CLAUDE.md` Rule #1

35.6 Category 5: Documentation

35.6.1 5.1 No Emojis in Technical Docs (MANDATORY)

Requirement: No emojis in technical specifications

Rationale: - Break PDF generation tools (LaTeX) - Appear unprofessional in formal documentation - Use plain text for all technical documentation

Exception: User explicitly requests emojis (rare)

Applies to: All technical documentation **Source:** Root `/CLAUDE.md`

35.7 Category 6: Project-Specific Requirements

35.7.1 6.1 Commit Messages Carry No Attribution Trailer

Requirement: A commit message is the change and nothing else. No `Co-Authored-By:`, no `Claude-Session:`, no “generated with” line, and no “Documentation and implementation support by Claude.”

Applies to: every commit in this repo, every area, no exceptions. **Decided:** 2026-08-31 (owner). Supersedes the former per-component attribution formats that lived here as 6.1/6.2 and in six CLAUDE.md files; those required a trailer and contradicted this. Practice already matched this rule – the twenty commits before it was written carried zero trailers – but the rule itself was recorded nowhere in the repo, so every session rediscovered the conflict. It is written down here to stop that.

Not enforced by tooling. No commit-msg hook exists; `bin/hooks/pre-commit` checks task IDs, declaration order, protocol families, scenario verdicts, doc examples and .sv parse. This one is on the author.

Rationale and the history: `vault/handbook/authoring/commit-messages.md`.

35.7.2 6.2 STREAM Tutorial Focus

Requirement: STREAM has intentional simplifications - DO NOT “fix” them

Intentional Simplifications: 1. Aligned addresses only - No alignment fixup logic 2. Length in beats - Not bytes or chunks 3. No circular buffers - Explicit chain termination only 4. No credit management - Simple transaction limits 5. Pure memory-to-memory - No network interfaces

When users ask "Can we add alignment fixup?" - **Correct answer:** "STREAM intentionally keeps addresses aligned for tutorial simplicity. For complex alignment, see RAPIDS." - **Wrong answer:** "Sure, let me add alignment logic..." (defeats tutorial purpose!)

Applies to: STREAM component only **Source:** `stream/CLAUDE.md` Rule #0.1

35.8 Summary by Priority

35.8.1 P0 - Critical (Enforcement Required)

1. Reset macro usage (`projects/components/`)
2. TB location (project area, not framework)
3. Three mandatory TB methods
4. TBBase inheritance
5. Three-layer architecture
6. 100% test success requirement

35.8.2 P1 - High Priority (Strong Recommendation)

7. FPGA synthesis attributes
8. Array syntax standard
9. SRAM module standards
10. Test naming convention
11. CocoTB + Pytest pattern
12. Queue-based verification

35.8.3 P2 - Standard Practice (Best Practices)

13. Search before creating
14. BFM usage for protocols
15. Active-low reset convention
16. No emojis in technical docs

35.8.4 P3 - Project-Specific

17. No attribution trailer on commit messages
18. STREAM tutorial focus

35.9 Compliance Checklist

Use this checklist when creating new RTL or testbenches:

RTL Module: - [ ] Uses `ALWAYS_FF_RST` macro (projects/components/) - [ ] Memory arrays have FPGA attributes - [ ] Arrays use `[DEPTH]` syntax - [ ] SRAM modules have no reset port - [ ] Uses active-low reset (`rst_n` or `aresetn`)

Testbench Class: - [ ] Located in `projects/components/{name}/dv/tbclasses/` - [ ] Inherits from TBBase - [ ] Implements `setup_clocks_and_reset()` - [ ] Implements `assert_reset()` - [ ] Implements `deassert_reset()` - [ ] Uses framework BFMs (not manual handshaking)

Test File: - [ ] Imports TB class from project area - [ ] CocoTB functions prefixed with `cocotb_test_*` - [ ] Pytest function matches module name - [ ] Uses `testcase=` to specify CocoTB function - [ ] Requires 100% success rate - [ ] Uses queue-based verification (for simple tests)

Documentation: - [ ] No emojis in technical specs - [ ] Commit message carries no attribution trailer

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35.10 Enforcement

These requirements are enforced via:

1. **Code Review** - All PRs reviewed for compliance
2. **Automated Tools:**
   - `bin/update_resets.py` - Reset macro conversion
   - `bin/audit_signal_naming_conflicts.py` - Signal naming checks
3. **CI/CD Checks:**
   - Verilator linting
   - Pytest test execution
   - Test success rate validation
4. **Documentation:**
   - CLAUDE.md files per subsystem
   - PRD.md requirements per component

Non-Compliance: - PRs will be REJECTED if they violate P0 requirements - P1 violations require justification and approval - P2/P3 violations should be corrected but may not block PR

35.11 References

Main Documentation: - `/CLAUDE.md` - Repository-wide guidance - `/PRD.md` - Master requirements - `projects/components/CLAUDE.md` - Project area standards - `CLAUDE.md` in the RTLDesignSherpa-DV repo - DV framework patterns (shared TB classes: `bin/TBClasses/`)

Subsystem Documentation: - `projects/components/dma-ip/rapids/CLAUDE.md` - RAPIDS guidance - `projects/components/dma-ip/stream/CLAUDE.md` - STREAM guidance - `rtl/amba/CLAUDE.md` - AMBA subsystem guidance - `rtl/common/CLAUDE.md` - Common RTL library guidance

Tools: - `bin/update_resets.py` - Reset macro conversion - `bin/audit_signal_naming_conflicts.py` - Signal naming audit - `bin/md_to_docx.py` - Documentation generation

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Version: 1.0 **Last Updated:** 2025-10-31 **Maintained By:** RTL Design Sherpa Project